Informality, Heterogeneity, and the Business Cycle

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Abstract

This paper develops a quantitative framework to analyze how informal labor markets shape employment dynamics over the business cycle in developing economies. I construct a search-and-matching model with heterogeneous workers, endogenous vacancy creation in formal and informal sectors, and aggregate productivity shocks. The model's block-recursive structure provides a tractable way to incorporate both worker heterogeneity and aggregate uncertainty. Calibrated to the Brazilian labor market, the model reproduces key qualitative patterns: informality and unemployment are countercyclical, low-skilled workers are more exposed to informal employment, and informal jobs are easier to access but less stable than formal jobs. The quantitative exercise highlights an equilibrium trade-off between search access and job stability. Informal submarkets expand the set of opportunities available to unemployed workers, especially when formal vacancy creation is weak. At the same time, they expose workers to higher separation risk. In a counterfactual economy that shuts down informal vacancy creation, workers lose this adjustment margin, but employment shifts toward more stable formal matches. Under the current calibration, the stability margin dominates: aggregate unemployment falls modestly and low-skilled welfare rises. These results emphasize that the macroeconomic role of informality cannot be summarized solely by its effect on employment levels. Equally important is how informality affects job-finding prospects, match survival, and the distribution of gains across workers over the cycle.

Keywords: Informal Sector; Cyclical Unemployment; Labor Market Institutions, Labor Market Policies, Regulations

JEL Codes: E24; E26; J08

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1 Introduction

Labor market informality is a defining characteristic of developing economies, with informal employment accounting for a substantial share of total employment. While often viewed as a symptom of institutional and regulatory failures, the informal sector plays a complex dual role. Although avoiding regulations undermines government tax revenue and compromises worker rights (as informality implies evading direct costs, such as taxes and social security contributions), it also means bypassing the operational rigidities (like hiring and firing rules) that can limit flexibility to changing economic conditions.

The state of the business cycle can crucially impact the decision process of firms and workers regarding informality, as both firms' incentives to create formal jobs and workers' valuation of informal employment opportunities vary with the aggregate conditions of the economy. This feature, however, is often overlooked in the literature. Despite the empirical evidence on the counter-cyclical behavior of informality and the differences in the patterns of the flows to and from unemployment of workers with distinct formality statuses, much of the previous work has focused on steady-state economies with no space for changing aggregate productivity.

This paper develops a quantitative framework to analyze how the presence of an informal sector shapes labor market dynamics over the business cycle. The model allows us to focus on two aspects: (1) the aggregate implications of informality for labor market outcomes during economic fluctuations, and (2) the distributional effects across workers with different skill levels.

Our framework displays heterogeneous workers who differ in their human capital and firms that can choose between creating formal or informal jobs. Hence, we focus on informality as informal wage work hired by firms rather than a frictionless self-employment state of the workers. Aggregate productivity shocks generate business cycle fluctuations, endogenous separations, and changes in the optimal formality sector choice of workers and firms.

Our methodological contribution lies in developing a tractable framework while incorporating worker heterogeneity, search and matching frictions, and aggregate uncertainty. The key to achieving this is the model's block recursive structure (Menzio and Shi, 2010, 2011; Kaas, 2023). This feature ensures that agents' value and policy functions depend on the aggregate state only through the level of aggregate productivity, eliminating the dependency on the infinite-dimensional distribution of workers across states. Block recursivity emerges as a combination of (i) directed search, in the sense that firms post vacancies in specific submarkets and workers self-select into their preferred locations, (ii) free entry of firms, and (iii) a constant returns to scale matching function. These three elements allow the market tightness of each submarket to be determined independently of the workers' distribution, and it thus compresses all information necessary for the worker's choice. This contrasts with random search models, where firms and workers face uncertainty about their potential matches, yielding value functions dependent on the full distribution of worker types and states.

We calibrate the model to match key qualitative patterns of the Brazilian labor market and then simulate it to study how informality shapes equilibrium vacancy creation, worker search choices, separations, and welfare over the cycle. The quantitative exercise highlights a central trade-off. Informal submarkets expand the set of search opportunities available to unemployed workers, especially for low-skilled workers and in aggregate states in which formal hiring is weak. At the same time, informal matches are less stable than formal ones. A counterfactual economy without informal vacancies makes this mechanism explicit: removing informality weakens search access but shifts employment toward more stable formal matches. Under the current calibration, the stability margin is strong enough to lower unemployment and raise average welfare for low-skilled workers, while high-skilled workers experience welfare losses. These results emphasize that the role of informality cannot be summarized only by its effect on employment levels; it also depends on how it changes job-finding prospects, match survival, and the distribution of gains across workers.

We build on a rich body of work analyzing labor market informality in steady-state environments. This literature has enhanced our understanding of the incentives faced by firms and workers and their implications on firm dynamics, productivity, and earnings, while also assessing the impacts of various policy and regulatory changes (Haanwinckel and Soares, 2021; Machado Parente, Brotherhood, and Iachan, 2025; Meghir, Narita, and Robin, 2015; Ulyssea, 2018, 2020). However, they abstract from the role of business cycle fluctuations.

The cyclical properties of informal labor markets are the focus of a relatively smaller literature. Shapiro (2014) studies the impact of having a large self-employed "sector" on the behavior of employment and output along the business cycle. Fiess, Fugazza, and Maloney (2010) also treat the informal sector as self-employment to understand its impact on the transmission of economic shocks. Fernández and Meza (2015), Leyva and Urrutia (2020), and Horvath and Yang (2022) build business cycle models with informal sectors and measure the impact of informality and labor regulation on macroeconomic volatility and the propagation of shocks. The first two also model informality as a frictionless self-employment state. Thus, they abstract from the intensive margin of informality, through which formal firms hire workers off the books. Closer to our framework is Bosch and Esteban-Pretel (2012). However, as in the previously mentioned papers, they model workers as ex-ante homogeneous. The block recursive setting of our framework allows us to study the distributional effects of the interaction of the business cycle and informality, as well as the impact of policy changes, by incorporating worker heterogeneity. We build on the search-and-matching tradition (e.g., Shimer (2005)), where aggregate productivity shocks interact with search frictions to drive labor market dynamics. On the firm's side, we endogenize both the job-finding and the separation rates, as well as the intensive margin of informality, by allowing firms to optimally choose when to create and terminate vacancies in both sectors.

In what follows, we start by presenting a brief empirical overview of the cyclical behavior of labor markets in Brazil (Section 2). Section 3 describes the model. Section 4 presents the parametrization and baseline dynamics, while Section 5 studies the no-informality counterfactual. Section 6 concludes.

2 Empirical Evidence

In this section, we provide empirical evidence on the cyclical patterns of informality in developing economies using Brazil as our setting. We document three empirical regularities that motivate the key features of our model: the counter-cyclical nature of informality, heterogeneity in labor market outcomes across skill groups, and wage differences across sectors.

We use microdata from *Pesquisa Mensal de Emprego Nova* (PME), a rotating household survey with monthly visits that covered urban areas of six large metropolitan regions. Its data is available for the period from 2002.03 to 2016.02.[1] We restrict our sample to prime-age workers (18-59 years). We adopt the methodology implemented by Data Zoom (2025) to explore the panel aspect of the microdata and construct indicators of the labor market transitions experienced by each worker over subsequent interviews.

Besides the fact that informality is a widespread phenomenon in the country, Brazil is an adequate setting for studying the topic because its institutional design provides a clear and widely understood notion of what constitutes a formal job: labor regulation requires that, when formally hiring a worker, the firm sign a document called *Carteira de Trabalho e Previdência Social* (CTPS). By doing so, the firm informs competent authorities of the hiring, committing itself to paying all mandatory benefits to the worker (such as compliance with the minimum wage and contributions to social security and a fund available upon layoff) and associated payroll taxes. Workers are aware of whether they were hired under a signed CTPS, and the survey design explicitly asks employed individuals if their work regime follows this formalized scheme or not. Thus, this provides a clear and effective way to classify the workers in the sample into the formal or the informal sector.

Workers in the sample belong to one out of seven employment statuses. The model we present in the next section includes three: employed in the private formal sector (those with a CTPS contract), employed as an informal wage worker (those working for someone else without a CTPS contract), and unemployed. The remaining statuses, excluded from the model, are self-employed, employers, public sector/military workers, and those out of the labor force.

Table 1 presents the cyclical properties of key labor market stocks (in logs). It reports the standard deviation and the correlation with economic activity of employment rates as a share of the working-age population, the unemployment rate among the private sector labor force, and the informality rate as a share of total private employment.[2] All series are seasonally adjusted

[1]We acknowledge two primary limitations of using PME: its discontinuation in 2016 and its restricted coverage to six metropolitan regions. Despite the availability of the more recent, nationally representative *Pesquisa Nacional por Amostra de Domicílios Contínua* (PNADC), we find PME better suited to our focus on business cycle fluctuations. PME offers a higher frequency than PNADC (monthly vs quarterly), has a longer temporal span, and covers three distinct recessionary periods (2003, 2008-2009, and 2014-2016). Conversely, PNADC only captures the 2014-2016 crisis and the COVID-19 recession, with the latter having significant analytical complications due to data collection disruptions and its highly atypical nature, making it difficult to compare with standard business cycles. We confirm, however, that the primary patterns documented here are also present in the PNADC data, and these results are available upon request.

[2]We treat the private sector as the combination of formal private workers and informal wage workers. The private labor force adds the unemployed to this last group. These definitions are consistent with the employment conditions of our model.

using the X-13 ARIMA-SEATS method and further smoothed using a centered moving average of adjacent periods to minimize high-frequency noise.

The data is in line with previous findings of the literature on counter-cyclical behavior of the informality rate (Bosch & Esteban-Pretel, 2012; Bosch & Maloney, 2008; Loayza & Rigolini, 2011; Roldos et al., 2019). While total and formal employment show pro-cyclical behavior, informal employment's cyclicality has the opposite sign, indicating that informal hiring can expand relative to formal hiring when aggregate conditions deteriorate. This pattern is often summarized as a *buffer* margin; in our sample it is primarily driven by informal wage work rather than self-employment. As we show when turning to flows, such a margin naturally coexists with much higher job instability in informal employment than in formal jobs. In terms of variance over time, formal employment shows a somewhat more stable trajectory than informal wage work. These patterns are consistent across skill groups: both low and high-skilled workers experience counter-cyclical informality rates and pro-cyclical formal employment, with similar magnitudes (see Table 2). The evidence suggests that the aggregate patterns thus reflect a systematic feature of the labor market rather than compositional changes.

Table 1: Labor Market Stocks

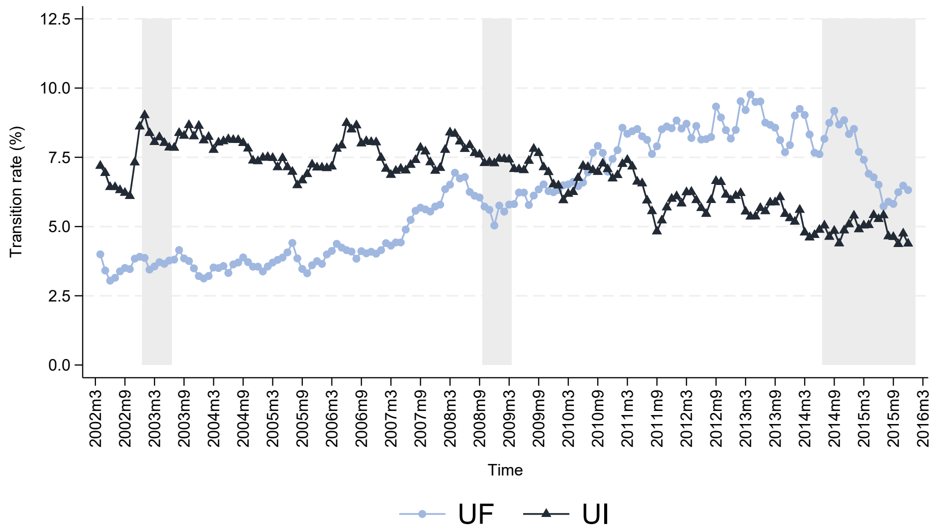
Variable	S.D.	Corr. with activity
Unemployment (% private labor force)	0.299	-0.427
Informality (% private employed)	0.221	-0.301
Employment (% work. age pop.)	0.037	0.360
Formal private employment (% work. age pop.)	0.123	0.313
Informal wage employment (% work. age pop.)	0.178	-0.275
Self-employment (% work. age pop.)	0.024	-0.099

Notes: logs of seasonally adjusted stocks by the X-13 ARIMA-SEATS method and further smoothed by a centered-moving average using the two adjacent periods. Stocks calculated using the surveys' expansion weights. Monthly activity measure is the cycle component of the smoothed and seasonally adjusted log of the IBC-BR index from Banco Central do Brasil (2025).

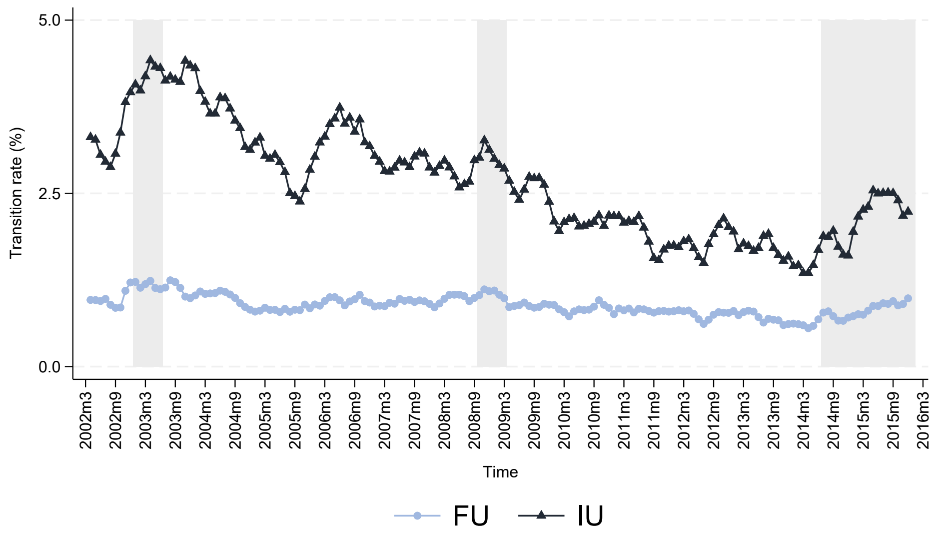
We now turn to the analysis of labor market flows. Figure 1 presents monthly transition rates between employment states over the 2002-2016 horizon of PME, a period that provides a comprehensive view of labor market dynamics across three recession and expansion cycles. Panel (a) shows job-finding rates from unemployment into formal employment (UF) and into informal wage employment (UI). Particularly during the 2014-2016 recession, the formal sector job-finding rate exhibits a substantial decline, while transitions into informal jobs remain relatively stable.

Figure 1: Cyclicality of labor market flows

(a) Job-finding rate



(b) Job-separation rate



Notes: monthly flows from PME, seasonally adjusted by the X-13 ARIMA-SEATS method and further smoothed by a centered-moving average using the two adjacent periods. Flows calculated using the survey's expansion weights. Shaded areas indicate recession periods as defined by CODACE (2023). "U" stands for unemployment, "F" for formal employment, "I" for informal wage work, and "S" for self-employment.

Panel (b), which displays job-separation rates (FU, IU), reveals important differences in job stability. The formal sector is characterized by consistently lower and more stable separation rates, averaging just 0.88% per month with only modest increases during downturns. The informal sector, in contrast, exhibits much higher instability: its separation rate (IU) averages 2.65% per month (nearly three times the formal rate) and rises significantly during recessions. This heightened cyclicality of the informal separation rate aligns with the findings of Gomes, Iachan, and Santos (2020), who also document that flows from informal employment to unemployment are more pronounced during recessions – a finding they obtain using the PNADC survey over a more recent period. This difference in cyclicality is likely a result of the firing costs imposed by formal sector regulations, which are absent in informal work. We capture this distinction in our model through sector-specific firing costs and exogenous separation rates.

These patterns suggest that the counter-cyclical nature of informality stems primarily from changes in job creation rather than job destruction. During economic downturns, while formal job opportunities become scarce, informal hiring from unemployment can remain comparatively resilient. At the same time, informal jobs exhibit higher separation rates than formal jobs, so any "buffer" interpretation must be understood jointly with high turnover on informal matches. Our quantitative analysis in Sections 4 and 5 uses the model to discipline how this tension between *search access* and *job stability* shapes unemployment, wages, and welfare when the informal sector is present versus shut down.

Table 2 summarizes key labor market outcomes by skill level, defined by high school completion (high-skilled) or non-completion (low-skilled). We use this criterion to map our model's skill groups to the data. Unemployment rates exhibit similar average levels, dispersion, and cyclical behavior across both worker types. This similarity also extends to separation rates.

Table 2: Labor Market Outcomes by Skill Level

Variable	Mean	S.D.	Corr. with activity
Low-skilled workers			
Unemployment rate	11.67	3.62	-0.367
Informality rate	33.88	4.65	-0.325
Job-finding: formal (UF)	5.38	2.07	0.379
Job-finding: informal (UI)	8.34	1.10	-0.214
Separation: formal (FU)	0.85	0.14	-0.367
Separation: informal (IU)	2.62	0.81	-0.361
High-skilled workers			
Unemployment rate	11.45	3.14	-0.405
Informality rate	19.04	4.11	-0.282
Job-finding: formal (UF)	6.52	1.94	0.424
Job-finding: informal (UI)	5.62	1.05	-0.252
Separation: formal (FU)	0.90	0.16	-0.413
Separation: informal (IU)	2.70	0.83	-0.348

Notes: Low-skilled = less than high school completion; High-skilled = high school degree or more. Rates in percentages. All series seasonally adjusted using X-13 ARIMA-SEATS method and further smoothed by a centered-moving average using the two adjacent periods. Monthly activity measure is the cycle component of the smoothed and seasonally adjusted log of the IBC-BR index.

In contrast, informality rates are markedly higher among low-skilled workers, averaging almost 34% in the period, versus only 19% among high-skilled workers. The job-finding rates reveal the source of this difference: high-skilled workers are more likely to find a formal job, while low-skilled workers face a higher probability of finding informal employment than formal employment. Importantly, the cyclical properties are qualitatively similar across skill groups—both experience counter-cyclical unemployment and informality. This suggests that aggregate shocks affect the entire labor market, albeit with potentially different intensities.

While the aggregate stocks and flows define the market's dynamics, the wage structure reveals the underlying incentives for workers. Table 3 presents key statistics on the wage structure by sector and skill. Panel A reports average real hourly wages. Unsurprisingly, wages are higher for high-skilled workers and for those employed in the formal sector.

Table 3: Wage Structure

Panel A: Average hourly wages		
	Low-Skilled	High-Skilled
Formal sector	10.45	20.84
Informal sector	8.39	16.67
Panel B: Sector and skill premiums		
Formal premium (log points)	0.0730	
High-skill premium (log points)	0.5255	

Notes: Panel A: wages in BRL from December 2024. Panel B: Formal premium from worker fixed effects regression controlling for age, age squared, and industry fixed effects. High-skill premium from pooled OLS controlling for age, age squared, gender, formality status, and metropolitan area and industry fixed effects. See full regression results on Table C3 in Appendix C.

However, simple wage averages are insufficient as they are skewed by significant selection effects — for example, higher-ability workers may disproportionately sort into formal jobs. In Panel B, we show our estimations of the sector and skill wage premiums.

To isolate the formal wage premium from time-invariant worker characteristics (including unobserved factors such as innate ability or motivation), we use an individual fixed-effects (FE) model on the panel of log real hourly wages. The specification also includes controls for age, its square, and industry fixed effects. The purpose of the latter is to remove wage differences that arise from workers sorting across industries, a margin that is absent from the model. Therefore, the moments should be interpreted as the within-worker, within-industry wage gap associated with formal employment.

The empirical specification is given by:

$$\ln(w_{it}) = \alpha_i + \beta_{formal}\,\text{Formal}_{it} + \gamma_1 \text{age}_{it} + \gamma_2 \text{age}_{it}^2 + \lambda_{\text{ind}(it)} + \epsilon_{it},$$

where α_i denotes worker fixed effects and $\lambda_{\text{ind}(it)}$ denotes industry fixed effects, so the coefficient β_{formal} is identified from within-worker wage variation net of cross-industry wage differences.

The coefficient of interest, β_{formal}, captures the wage change a worker experiences when switching sectors, purged of their fixed personal attributes. We find a positive coefficient of 0.0730, which implies that, after controlling for worker-level characteristics, formal sector wages are 7.6% higher on average than informal wages.[3]

The fixed-effects model, by design, cannot estimate the premium for a time-invariant characteristic like skill. Therefore, to identify the skill wage premium, we estimate a separate pooled OLS Mincer regression of log real hourly wages on the high-skill dummy. This specification also includes a formal-sector indicator and additional controls, including industry fixed effects, so that the estimated coefficient captures the wage gap between skill groups net of cross-industry wage composition:

$$\ln(w_{it}) = \alpha + \beta_{skill}\,\text{HighSkill}_i + \mathbf{\Gamma}' X_{it} + \epsilon_{it},$$

where X_{it} includes a formal-sector indicator, age, age squared, gender, metropolitan area fixed effects, and industry fixed effects.

The resulting estimate of coefficient β_{skill} is 0.5255, indicating that high-skilled workers earn, on average, 69.1% more than their low-skilled counterparts after controlling for formality, industry, age, and demographics. Together, these patterns motivate our model's specification of higher productivity in the formal sector and skill-differentiated human capital.

3 The Model

In this section, we present a search and matching model of a labor market with formal and informal sectors. The model is designed to capture the key empirical patterns documented in the previous section, particularly the cyclical dynamics between formal and informal employment and the heterogeneity across human capital levels. The model features directed search by unemployed workers and free entry by firms, which endogenously determine the market tightness for all potential submarkets.

The economy is populated by a continuum of workers with latent human capital $h \in \mathcal{H} = \{h_1, \ldots, h_{N_h}\}$, where $\mathcal{H}$ is a fixed grid. There is a continuum of firms with positive measure. Each worker has an endowment of 1 (indivisible) unit of labor. Workers have linear instantaneous utility and all agents have discount factor β.[4]

There are two types of job relations ("sectors"), $s \in \{F, I\}$, where F stands for formal and I stands for informal. A job in the formal sector is subject to a minimum wage, a benefit for the worker in case of losing the job, and the application of costs: (i) wage taxes for the workers; (ii) payroll and profit taxes for the firms; and (iii) firing costs for the firms. An informal job is not

[3] The percentage is calculated as $(\exp(\hat{\beta}) - 1) \times 100$.

[4] Linear utility allows for a simple way to model unemployment insurance without introducing state-dependent utility functions.

subject to any of these labor market regulations. However, it is subject to a higher exogenous separation rate and uses a less productive technology.

There is an aggregate productivity shock z. The productivity of a job depends on the human capital level h of the worker and has an idiosyncratic time-fixed component, y, which is drawn after the match occurs from a distribution G. The production technology is sector-dependent, with $f(h, s, y; z) = \exp\{A_s + z + y\} \times h$, where $A_F > A_I$ captures the fact that formal firms have access to better institutions, such as credit markets, litigation services, etc. The vacancy posting cost in sector s is κ_s. Unemployed workers have home production $b(h)$.

Each worker searching for a job chooses a particular submarket indexed by $\eta = (h, s, w; z)$, where h is their human capital level, s is the sector, w is the wage offered, and z is the aggregate state of the economy. Note that, as human capital level h and the aggregate state z are given, the choice component of a submarket is (s, w). There is a constant return to scale matching function $M(u, v)$, where u is the number of unemployed workers and v is the number of vacancies. Each submarket has market tightness $\theta(\eta) = \frac{v(\eta)}{u(\eta)} \geq 0$, which is determined in equilibrium. The probability of a worker in a given market being selected to a match and the probability of a firm filling a vacancy posted in a market are, respectively:

$$p\big(\theta(\eta)\big) = \frac{M\big(u(\eta), v(\eta)\big)}{u(\eta)} = M\big(1, \theta(\eta)\big) \tag{1}$$

$$q\big(\theta(\eta)\big) = \frac{p\big(\theta(\eta)\big)}{\theta(\eta)}. \tag{2}$$

For each period t, the timing of the economy is as follows:

i) The aggregate uncertainty is resolved.
ii) Separations happen exogenously and as an endogenous choice of the firms.
iii) Unemployed workers search, deciding which market to visit.
iv) Matches occur and productivity levels are drawn for the newly formed jobs.
v) Production, consumption, and tax payments occur.

This timing implies a sharp distinction between unemployment and employment spells in simulated data: a worker who remains employed never reallocates across submarkets within the period, and (given the absence of on-the-job search) cannot move directly from one match to another without an intervening unemployment spell.

3.1 Bellman Equations

Unemployed worker An unemployed worker with human capital h enjoys utility from their home production level $b(h)$ and chooses the sector s and wage w that characterize a submarket $\eta = (h, s, w; z)$ to search for a job, given aggregate productivity level z. They are matched to a vacancy with probability $p(\theta(\eta))$. If the match happens, the productivity y is drawn from G and they become an employed worker with value function $W(h, s, w, y; z')$ next period. We denote by $m^U(h;z) \in (S \times W)$ the policy function of the optimal submarket choice of the unemployed worker. The value function for an unemployed worker is

$$U(h;z) = b(h) + \beta\, \mathbb{E}_{z'}\left[S^U(h;z') \,\middle|\, z\right], \tag{3}$$

where $S^U(h;z)$ is the value of searching while unemployed for a worker of human capital level h when the aggregate state is z, given by

$$S^U(h;z) = \max_{s,w}\left\{U(h;z) + p\big(\theta(h,s,w;z)\big)\left(\mathbb{E}_y\Big[W(h,s,w,y;z)\Big] - U(h;z)\right)\right\}. \tag{4}$$

Employed worker

Denote by $x = (h,s,w,y;z)$ the state of a worker with human capital h employed in sector s, with wage w, and idiosyncratic productivity y when the aggregate productivity level is z. Their value function is $W(x)$. Over their wage w, the worker is subject to a tax rate $\tau^w(s)$ given by

$$\tau^w(s) = \begin{cases} \tau^w > 0 & \text{if } s = F \\ 0 & \text{if } s = I, \end{cases} \tag{5}$$

that is, there is income tax only in the formal sector.

An ongoing match is subject to endogenous separation by the firm. If the firm wants to keep the match, there is still an exogenous separation rate $\delta(s)$. Let $d(x)$ represent this consolidated separation rate, which we define more precisely later in this section. If the match ends, the worker is entitled to receive benefits $B(s) \times w$, where the rate $B(s)$ is given by

$$B(s) = \begin{cases} B > 0 & \text{if } s = F \\ 0 & \text{if } s = I. \end{cases} \tag{6}$$

This captures the fact that only formal jobs are associated with unemployment insurance, which we model as a one-time payment upon separation. A separated worker can search immediately as an unemployed worker.

Let $x' = (h,s,w,y;z')$. The value of an ongoing match for the worker with state x is

$$W(x) = \big(1 - \tau^w(s)\big)w + \beta\mathbb{E}_{z'}\left[W(x') + d(x')\Big(B(s)w + S^U(h;z') - W(x')\Big) \,\middle|\, z\right]. \tag{7}$$

Note that the only state variable that changes between periods for an employed worker is the aggregate productivity level z. This change might trigger a separation decision by the firm.

Value of a vacancy

A vacancy posted by a firm in the submarket $\eta = (h,s,w;z)$ costs κ_s to be maintained open and has an associated value of

$$V(\eta) = -\kappa_s + q\big(\theta(\eta)\big)\mathbb{E}_y\left[J(h, s, w, y; z)\right]. \tag{8}$$

The expected value over y is present because the idiosyncratic productivity y is drawn from G after the match is formed, yielding value $J(h, s, w, y; z)$ for the firm.

With free entry, it must be the case that

$$\kappa_s \geq q\big(\theta(\eta)\big)\mathbb{E}_y\left[J(h, s, w, y; z)\right] \tag{9}$$

and $\theta(\eta) \geq 0$ with complementary slackness. That means that (9) holds with equality ($V(\eta) = 0$) in every submarket where firms post vacancies (markets where $\theta(\eta) > 0$). Hence, the equilibrium condition for vacancy posting can be rewritten as

$$\theta(\eta) = q^{-1}\left(\frac{\kappa_s}{\mathbb{E}_y\left[J(h, s, w, y; z)\right]}\right) \quad \text{if } \theta(\eta) > 0. \tag{10}$$

Value of a match to the firm

When a match of type $x = (h, s, w, y; z)$ is maintained, the firm enjoys a profit flow $\pi(x)$ given by

$$\pi\big(x\big) = \big(1 - \tau^p(s)\big)\Big(e^{A_s+z+y} \times h \;-\; \big(1 + \tau^f(s)\big)w\Big) \tag{11}$$

The firm is subject to sector-specific payroll taxes and profit taxes at rates given by

$$\tau^f(s) = \begin{cases} \tau^f & \text{if } s = F \\ 0 & \text{if } s = I \end{cases} \tag{12}$$

and

$$\tau^p(s) = \begin{cases} \tau^p & \text{if } s = F \\ 0 & \text{if } s = I. \end{cases} \tag{13}$$

After the aggregate uncertainty is resolved, the firm can choose to terminate a no longer profitable match. The match is terminated with probability one if the continuation value for the firm is inferior to the cost of separation, $C(s)$, with

$$C(s) = \begin{cases} C > 0 & \text{if } s = F \\ 0 & \text{if } s = I. \end{cases} \tag{14}$$

Otherwise, the match ends accordingly to the exogenous separation rate $\delta(s)$ – which might differ by sector:

$$\delta(s) = \begin{cases} \delta_F \geq 0 & \text{if } s = F \\ \delta_I > 0 & \text{if } s = I. \end{cases} \tag{15}$$

The value for the firm of an ongoing match $x = (h, s, w, y; z)$ is thus given by

$$\begin{aligned} J(x) &= \pi(x) + \beta \mathbb{E}_{z'} \left[\max_{d} \Big\{ d\big(-C(s)\big) + (1-d) J(x') \Big\} \Big| z \right] \\ &\text{s.t.} \quad \delta(s) \leq d \leq 1. \end{aligned} \tag{16}$$

3.2 Distribution of workers across states

Let $\mu_z(a)$ denote the measure of employed workers in a match of type $a = (h, s, w, y)$ when the aggregate state is z and $u_z(h)$ be the measure of unemployed workers over state $(h; z)$, both at the production stage. In a given instant, the inflow of people into unemployment is formed by the employed workers who were laid off and were unsuccessful in their search attempt:

$$\sum_a \Big[\mu_z(a) \times d(a; z) \big(1 - p\big(\theta(h, m^*; z')\big)\big) \Big], \tag{17}$$

where $m^* = m^U(h; z')$.

On the other hand, the outflow from unemployment is composed of the individuals who were matched to a firm and drew high enough productivity to sustain the match in their submarket of choice:

$$\sum_h \Big(u_z(h) \times p\big(\theta(h, m^*; z')\big) \Big). \tag{18}$$

3.3 The government

The government has four sources of revenue:

i) Income tax at rate τ^w charged only of workers in formal matches:

$$\tau^w \times \sum_{h,w,y} \Big(w \times \mu_z(h, F, w, y) \Big)$$

ii) Payroll tax at rate τ^f charged only of firms in formal matches:

$$\tau^f \times \sum_{h,w,y} \Big(w \times \mu_z(h, F, w, y) \Big)$$

iii) Profit tax at rate τ^p charged only of firms in formal matches:

$$\tau^p \times \sum_{h,w,y} \Big[\Big(e^{A_s + z + y} \times h - \big(1 + \tau^f(s)\big) w \Big) \times \mu_z(h, F, w, y) \Big]$$

iv) Firing costs C charged of formal matches subject to separation:

$$C \times \sum_{h,w,y} \Big(\mu_z(h, F, w, y) \times d\big(h, F, w, y; z'\big) \Big),$$

where the timing should be noted: $\mu_z(x)$ is the measure of workers in matches x at the production stage when the aggregate state is z. Next period, the aggregate state is z', which leads to firing decisions ruled by $d(x; z')$. Hence, at tomorrow's firing stage, the measure of workers with state x is still $\mu_z(x)$.
On the expenditure side, the government provides unemployment insurance payments B to workers separated from formal matches according to the total separation rule $d(\cdot)$. The amount paid by the government in benefits (in the production stage next period) is

$$B \times \sum_{h,w,y} \Big(w \times \mu_z(h, F, w, y) \times d\big(h, F, w, y; z'\big) \Big)$$

Block recursivity requires that we relax the requirement of a balanced government at every period.

3.4 Equilibrium

Let $x = (h, s, w, y; z)$ represent a match state and $\eta = (h, s, w; z)$ represent a submarket. A recursive equilibrium in this economy is a set of individual policy functions for unemployed search $m^U(h; z)$, firing by the firms $d(x)$, market tightness functions $\theta(\eta)$, and distributions $u_z(h)$ and $\mu_z(h, s, w, y)$ for the unemployed and employed workers over their states, respectively, such that:

1. Agents' decision rules are optimal.
2. The labor market tightness satisfies the free entry condition given in Equation 10.
3. The distribution of workers across states is consistent with individual policy functions.

Appendix A derives the existence and uniqueness of the model's block recursive equilibrium (BRE) using the Banach Fixed-Point Theorem. The key implication of the BRE is that the equilibrium objects in conditions (1) and (2) depend on the aggregate state only through z and not through the distributions of workers, a complicated multidimensional object. This property allows the "block" of value functions and *prices* (the market tightness) to be solved independently of the aggregate state distribution.

Furthermore, given the absence of on-the-job search and worker-driven separations, our model affords a further simplification: the firm block (value functions J and V, separation policy d, and market tightness θ) can be solved in a first stage, independently of the worker's value functions. This two-stage decoupling, which is not a required feature of all BRE models, significantly simplifies our proof and computational solution.

4 Calibration and Baseline Results

In this section, we describe how we calibrate the model to match key features of the Brazilian labor market and discuss the dynamics of our baseline economy. We start by defining the functional forms adopted in the model and then discuss the calibration of its parameters. Then, we move to our main quantitative analysis, which allows us to illustrate the role of the informal sector throughout the business cycle.

4.1 Parametrization

Table 4 summarizes the core functional forms used in the implemented model. Workers have linear utility over income. Home production is a linear function of latent human capital, $b(h) = \rho_b h$. Match output depends on the latent human capital of the worker, the productivity of the sector in which they work, the idiosyncratic productivity of the match, and the aggregate productivity level through an exponential specification. The latent human-capital distribution is discretized on a finite grid and linked to observed education labels only when moments are aggregated.

Table 4: Functional forms of the model

Function	Description	Functional Form	
$u(w)$	Utility function	Linear	$u(w) = w$
$b(h)$	Home production function	Linear	$b(h) = \rho_b h$
$f(\cdot)$	Production function	Exponential	$f(h, s, y, z) = e^{A_s + y + z} \times h$
$M(u, v)$	Matching function	DRW Function	$M(u, v) = \frac{u \cdot v}{(u^\alpha + v^\alpha)^{\frac{1}{\alpha}}}$
$G(y)$	Distribution of y	Normal	$y \sim N(\mu_y, \sigma_y^2)$
$\Phi(z)$	Law of motion of z	AR(1)	$z_t = \rho_z z_{t-1} + \epsilon_t$, where $\epsilon_t \sim N(0, \sigma_\epsilon^2)$

The constant-returns-to-scale matching function follows den Haan, Ramey, and Watson (2000), and implies that $p(\theta) = \theta\,(1 + \theta^\alpha)^{-\frac{1}{\alpha}}$. We assume the idiosyncratic productivity y is drawn from a normal distribution with mean μ_y and standard deviation σ_y. For the aggregate productivity level, we assume it follows a mean-zero AR(1), which we discretize using the Rouwenhorst method as a five-state grid: $Z = [z_1, z_2, z_3, z_4, z_5]$, where $z_1 < z_2 < 0$ are bad states (recessions), $z_3 = 0$ is a neutral state, and $0 < z_4 < z_5$ are good states.

Latent human capital and observable skill groups In the model, workers are heterogeneous in their latent human capital $h \in \mathcal{H} = \{h_1, \ldots, h_{N_h}\}$. In the data, however, we only observe their broad educational attainment. Using their high-school completion status, we classify workers in the data into two skill groups: low-skilled (L) and high-skilled (H), with π_H representing the fixed empirical share of high-skilled workers. To link the model's latent human-capital distribution to the observed skill groups, we use Bayesian weights to aggregate model-simulated outcomes into the observed skill labels moments. See Appendix B for details.

We assume latent human capital h follows a Log-Normal distribution for each demographic skill group $j \in \{L, H\}$:

$$\ln(h) \mid j \sim \mathcal{N}(\mu_j, \sigma_h^2)$$

The structural parameters governing these distributions are the common dispersion parameter (σ_h) and the two location parameters (μ_L and μ_H). We fix the low-skilled group location parameter by normalization such that $\mathbb{E}[h \mid L] = 1$. This implies $\mu_L = -\sigma_h^2/2$.

4.2 Calibration

The model's parameters are set in two groups. The first group consists of the externally calibrated parameters, whose values are drawn from institutional rules or calculated directly from the data (as detailed in Appendix C). This approach anchors the model to key real-world observations. They are presented in Table 5.

Table 5: Externally Calibrated Parameters

Symbol	Value	Description	Source / Rationale
A_I	0.0	Informal sector productivity	Model normalization
μ_L	−0.0029	Low-skilled human capital mean	Model normalization of $\mathbb{E}[h \mid L] = 1$
π_H	0.5415	Share of high-skilled workers	PME
β	0.99488	Discount factor	Set to match monthly real interest rate
α	0.2	Matching function elasticity	From Menzio & Shi (2010)
B	3.0	UI benefits multiplier	Statutory rules
τ^w	0.11	Income tax rate	Max. effective tax rate (Sindifisco Nacional, 2023)
τ^f	0.375	Payroll tax rate	Statutory values (Bosch & Esteban-Pretel, 2012; Ulyssea, 2018)
τ^p	0.0	Profit tax rate	Simplified out

The second group consists of internally calibrated parameters. Their values are chosen jointly to make the model's simulated behavior match key moments from the data. These parameters are presented in Table 6. We adopt a Simulated Method of Moments (SMM) approach, which we detail in Appendix C. The goal is to find a parameter vector that generates the qualitative cyclical patterns of the labor market, including a countercyclical informal sector that is more heavily populated by low-skilled workers, while also providing a reasonable quantitative fit to the targeted moments. The fit is summarized in Table 7.

Table 6: Internally Calibrated Parameters

Parameter	Description	Value
ρ_b	Home-production slope	0.383
μ_H	High-skilled location parameter	0.629
σ_h	Dispersion of latent human capital	0.076
A_F	Formal productivity	0.318
κ_F, κ_I	Vacancy costs	(0.225, 0.130)
δ_F, δ_I	Exogenous separation rates	(0.008, 0.014)
C	Firing costs	3.478
$\bar{m}_w$	Minimum wage	1.821
μ_y, σ_y	Idiosyncratic shock parameters	(1.078, 0.128)
$\rho_z, \sigma_\varepsilon$	Aggregate shock parameters	(0.923, 0.154)

The model successfully captures the core qualitative and cross-sectional patterns of the labor market. In line with the evidence in Section 2, it reproduces the key directional business-cycle co-movements of key labor-market aggregates: unemployment and informality are countercyclical, and formal job finding is procyclical. Furthermore, the model achieves a strong quantitative fit for several level moments, including unemployment rates, informality rates, the minimum wage ratio, and wage premiums across both the formal-informal and skill-level dimensions. In this sense, the framework is successful in capturing the main directional relationships emphasized in the data.

At the same time, important discrepancies remain. The model faces quantitative challenges regarding transition flows and cyclical dispersion. Specifically, it understates informal job-finding rates, overstates formal separations, and generates excessive volatility across both stocks and flows compared to empirical observations. The current calibration relies heavily on the aggregate shock process to generate cyclical movements, and the model often responds with sharp, non-linear reallocations across sectors when aggregate conditions cross specific state thresholds. In practice, this creates knife-edge transitions, translating into the observed high volatility of transition flows and stocks. Hence, this should be interpreted as a preliminary calibration, with further refinements in progress. Nonetheless, the model is already useful for identifying the mechanisms through which search behavior, separation risk, and sectoral reallocation jointly shape unemployment and informality over the cycle.

Table 7: Model Fit: Data vs. Model Moments

Moment	Data	Model
Key moments		
Unemployment rate, low-skilled	11.6671	13.3987
Unemployment rate, high-skilled	11.4529	11.9047
Informality rate, low-skilled	33.8790	36.3792
Informality rate, high-skilled	19.0439	22.2610
Correlation(Unemployment, Activity)	-0.5755	-0.7220
Correlation(Informality, Activity)	-0.3467	-0.3991
High- vs. low-skilled wage premium (log points)	0.5255	0.6520
Formal vs. informal wage premium (log points)	0.0730	0.0910
Minimum Wage Ratio	0.6110	0.6397
Std. Dev. Log Wages	0.6730	0.3764
Std. Dev. Activity (cycle)	0.0199	0.3264
Lag-1 Autocorrelation Activity (cycle)	0.9564	0.7521
Other moments		
Formal job-finding rate (UF), low-skilled	8.2200	5.3878
Formal job-finding rate (UF), high-skilled	9.4375	9.7728
Informal job-finding rate (UI), low-skilled	12.6513	4.5404
Informal job-finding rate (UI), high-skilled	8.0309	2.3919
Formal separation rate (FU), low-skilled	0.8840	1.2495
Formal separation rate (FU), high-skilled	0.9511	1.5373
Informal separation rate (IU), low-skilled	3.0896	1.9065
Informal separation rate (IU), high-skilled	3.2004	1.7817
Std. dev. unemployment rate	0.8972	5.0649
Std. dev. informality rate	0.6115	2.3624
Std. dev. UF rate	0.9470	2.4686
Std. dev. UI rate	0.8636	1.7651
Std. dev. FU rate	0.1029	2.9339
Std. dev. IU rate	0.4358	2.7374
Correlation(UF, Activity)	0.5580	0.7574
Correlation(UI, Activity)	0.0478	-0.5798
Correlation(FU, Activity)	-0.3326	-0.4089
Correlation(IU, Activity)	-0.3121	-0.4224

4.3 Simulation and Baseline Dynamics

To measure the dynamic role of the informal sector during changing economic conditions, we simulate an economy with $N = 5000$ individuals for 600 periods.[5] Each worker is assigned a fixed latent human-capital index drawn uniformly over the equal-mass grid $\mathcal{H}$. We obtain randomly generated trajectories for the aggregate productivity shock, the draws of the idiosyncratic productivity levels, the exogenous separation shocks, and the realization of the matching probabilities. We identify recession periods as moments when the aggregate productivity level

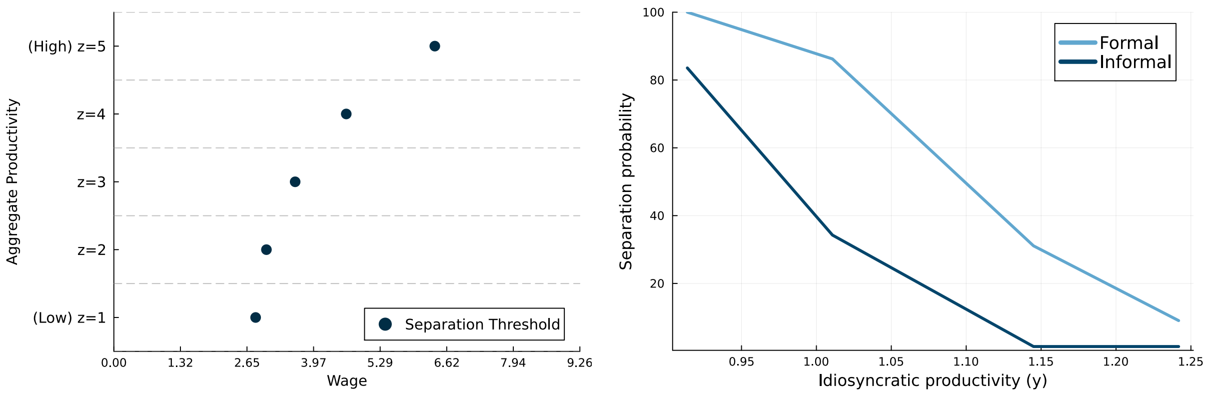
[5]More precisely, we simulate the model for 950 periods but drop the initial 350 as burn-in.

reaches negative values (z_1 and z_2), and "normal times" when it assumes the intermediate or positive levels. Moments split by low- and high-skill groups are then recovered from the simulated panel using Bayesian weights (see Appendices B and C).

The baseline model successfully reproduces the countercyclical behavior of the unemployment and informality rates observed in the data. This emerges from how endogenous choices by agents – vacancy creation and match destruction by firms, and search-location choices of unemployed workers – respond to the changing aggregate state. Recessions lead to an increase in endogenous separations by firms, as previously profitable matches are no longer sustainable given the worsened aggregate productivity level. This is illustrated in Panel (a) of Figure 2, which shows the endogenous separation threshold of a firm for different wages w and aggregate state levels z, in a match of type $(h = h_{\text{mid}}, s = F, y = y_{\text{mid}})$, where h_{mid} and y_{mid} denote the indices at the midpoint of the respective grids. For an active match, the wage is fixed. When the aggregate state changes, the separation threshold shifts and the fixed wage may no longer be profitable, triggering the firm to terminate the job relation. The specific threshold level depends on the other match characteristics, but the general pattern is that, during downturns, the set of wages the firm is willing to pay is reduced.

Figure 2: Firms' Separation Policy

(a) Endogenous Separation Thresholds (b) Average Separation Hazard in Recessions

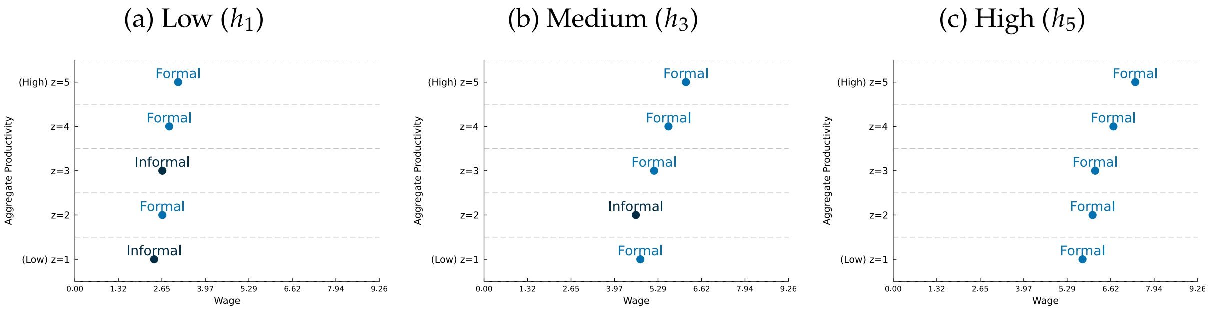


Notes: Panel (a) illustrates the endogenous separation thresholds for firms for the match type $(h = h_{\text{mid}}, s = F, y = y_{\text{mid}})$ across different aggregate states z. Panel (b) displays, for different idiosyncratic productivity levels y and separately by sector, the average separation hazard during recession periods.

Panel (b) of Figure 2 plots the policy-implied separation hazards during recessions. For each sector, the curves represent the separation rule $d(h, s, w, y, z)$ averaged across recessionary aggregate states (z) and all active submarkets (h, w) for each level of idiosyncratic productivity (y). As expected, hazards are higher for less productive matches, which are more likely to become unprofitable when aggregate conditions deteriorate. Notably, the formal sector exhibits higher separation hazards than the informal sector, likely reflecting the heavier burden of regulatory costs. This suggests a "cleansing effect" that operates unevenly across sectors: while formal firms aggressively terminate low-quality matches during downturns, the informal sector tends to retain them, potentially misallocating resources to less efficient employment ties. This pattern is *not* inconsistent with the data documenting much larger informal separation *rates* (IU) than formal ones (FU) in Section 2, as empirical IU and FU are transition rates integrating composition, exogenous risk, and endogenous destruction, whereas Panel (b) averages the optimal separation policy within each sector over active matches types.

Relatedly, unemployed workers switch the submarket they search in depending on the aggregate state of the economy. Figure 3 illustrates unemployed workers' policy functions for different levels of human capital (h) across aggregate productivity levels (z). Human capital plays an important role in determining the endogenous choice of labor market segment. For workers on the left tail of the latent human capital grid (h_1, Panel a), the optimal submarket choice involves frequent shifts between sectors: they target informal submarkets during both deep recessions ($z = 1$) and median aggregate states ($z = 3$). This suggests that the informal sector serves as a crucial entry point or a "fallback" option when formal vacancies are scarce.

Figure 3: Unemployed Workers' Policy Functions for Different Human Capital Levels



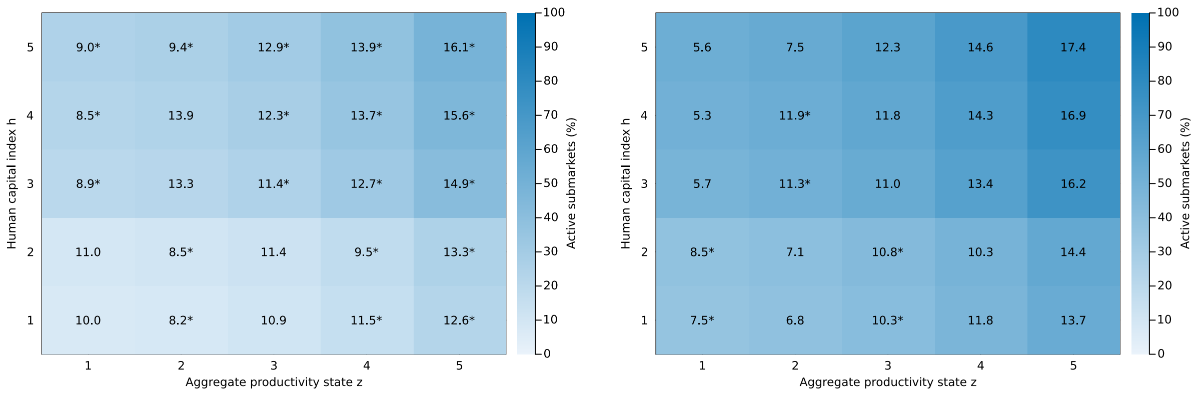
Notes: Each panel plots, for a different human capital level (h), the optimal search decisions for unemployed workers at different aggregate productivity states (z). Each dot represents the wage-sector submarket (w, s) that maximizes the worker's value of searching given (h, z). Wages are shown in the horizontal axis, while the sector associated with the chosen submarket is indicated by the labels "Formal" and "Informal" and by the dot color.

In contrast, workers with high human capital (h_5, Panel c) consistently target formal submarkets regardless of the aggregate state. The figure also reveals a clear "upgrading" effect in search behavior as aggregate conditions improve. Across all skill levels, higher aggregate productivity levels are associated with larger target wages, as seen by the rightward shift of the policy dots as z increases. During expansions, workers are more likely to search for higher-paying opportunities, while in recessions they target submarkets with lower wages but relatively higher probabilities of finding a job.

It is worth highlighting that the job-finding probabilities in different submarkets are not exogenous, but equilibrium objects resulting from the optimal search and matching efforts of workers and firms. Figure 4 shows that vacancy creation is strongly shaped by both aggregate conditions and worker heterogeneity. In both sectors, the set of submarkets with positive market tightness (and, thus, positive vacancy posting) expands as aggregate productivity improves, indicating procyclical firm entry. This expansion, however, is uneven across sectors. Formal vacancy creation is especially limited for low-human capital workers in bad states, becoming broader only for high-human capital workers. In contrast, the informal sector offers a wider set of active submarkets throughout the human capital distribution, making it a more accessible destination for low- and middle-human capital workers. The starred cells show that unemployed workers internalize these equilibrium vacancy-creation patterns when choosing where to search: when formal opportunities are scarce, especially in downturns, lower-human capital workers optimally redirect search toward informal submarkets, while higher-human capital workers remain attached to formal search.

Figure 4: Vacancy Creation and Job-Finding Probabilities by h and z

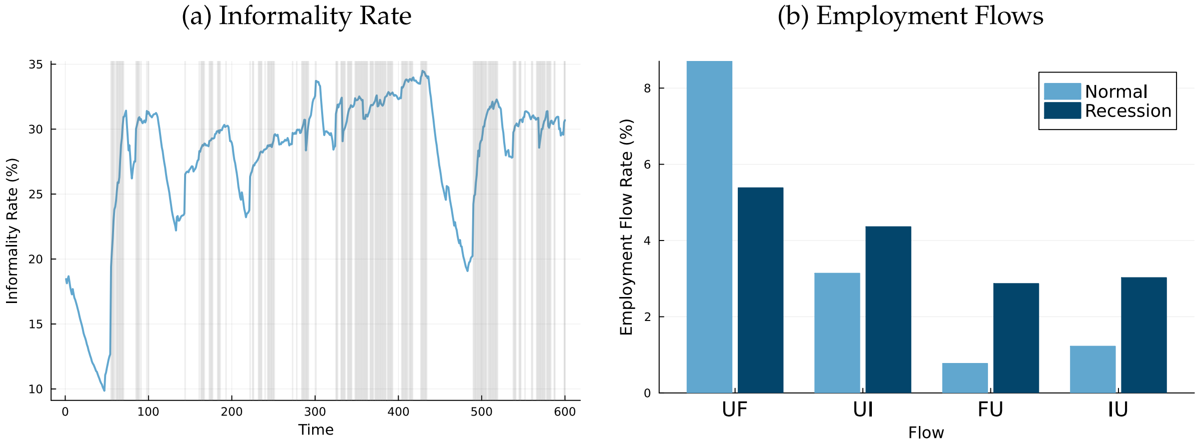
(a) Formal (b) Informal



Notes: Cell color reports the share of wage submarkets in which firms create vacancies, defined by positive equilibrium market tightness $\theta(h, s, w, z) > 0$. Cell entries report the job-finding probability, in percent, at the wage chosen by the worker in that aggregate state, evaluated separately in each sector. A star indicates the sector actually chosen by unemployed workers for that (h, z) state.

These endogenous search, match-separation, and vacancy-creation decisions are the micro-foundations for the aggregate labor market dynamics illustrated in Figure 5. Panel (a) plots the evolution of the simulated informality rate, which displays clear countercyclical behavior, spiking during the shaded recessionary periods. This macro-level volatility reflects the synchronized pressure of the margins discussed above: firms' increased separation of formal matches and the simultaneous shift in vacancy creation that leads workers' search toward informal submarkets. Consequently, the informal sector acts as a cyclical sponge, absorbing labor shed by the formal sector during contractions. As aggregate productivity recovers and the formal vacancy-creation margin reopens, the informality rate declines, though the simulation reveals significant persistence in the informality stock, suggesting that reallocating labor back into formal matches is a gradual process.

Figure 5: Aggregate Dynamics and Transitions Over the Cycle



Notes: This figure presents outcomes from the model's simulation. Panel (a) displays the evolution of the aggregate informality rate over time. Gray shaded areas indicate recession periods. Panel (b) reports the average transition rates between unemployment (U), formal employment (F), and informal employment (I) in normal versus recessionary periods. The flow rates are expressed as percentages of the origin state's population.

The mechanics behind these fluctuations are further clarified by the employment flows in Panel (b). During recessions, the formal job-finding flows (UF) drops significantly, while the informal job-finding flows (UI) increase mildly. Also consistent with the empirical evidence, separation rates in both sectors (FU and IU) are higher during downturns. The simulated *levels* of UI remain below those in the data (as reported in Table 7), so this within-simulation comparison of recession versus normal times should be read alongside the fit discussion above.

5 Counterfactual: Search Access and Job Stability without Informality

In this section, we conduct a counterfactual experiment to assess the role of the informal sector in shaping labor market outcomes. Specifically, we simulate a scenario in which firms cannot create informal vacancies, so their resulting market tightness is always zero, effectively shutting down the informal sector. The simulated random draws are the same as in the baseline economy to ensure comparability.

We start by comparing the baseline economy with the no-informality counterfactual in terms of key labor market outcomes, including unemployment rates, wages, and welfare. Table 8 presents these outcomes by worker group and aggregate state.

Table 8: Baseline Economy vs. No-Informality Counterfactual

	Baseline (A)		No Informality (B)		Difference (B-A)	
	Recession	Normal	Recession	Normal	Recession	Normal
All Workers						
Unemployment	17.461	9.862	16.752	9.314	-0.709	-0.548
Std. dev. of unemployment	7.096	3.957	6.573	4.100	-0.523	0.143
Average welfare	748.736	754.898	749.009	754.744	0.273	-0.154
Wage	4.440	4.531	4.497	4.580	0.057	0.049
Low-Skilled Workers						
Unemployment	18.086	10.912	16.731	10.468	-1.354	-0.444
Std. dev. of unemployment	7.358	4.131	5.986	4.183	-1.372	0.053
Average welfare	471.019	475.360	473.059	476.738	2.040	1.377
Wage	2.816	2.863	2.883	2.925	0.066	0.062
High-Skilled Workers						
Unemployment	17.052	9.174	16.766	8.558	-0.286	-0.616
Std. dev. of unemployment	7.047	3.918	7.042	4.119	-0.005	0.200
Average welfare	930.815	938.171	929.929	937.013	-0.886	-1.158
Wage	5.491	5.603	5.558	5.642	0.066	0.039

Notes: The table reports simulated averages by worker group and aggregate state. Unemployment statistics are reported in percentage points. The last two columns report the no-informality counterfactual (B) minus the baseline economy (A).

Table 8 shows that the no-informality counterfactual produces lower unemployment and higher average wages, despite removing a search option that workers actively choose in the baseline economy. For all workers, unemployment falls by 0.71 percentage points in recessions and by 0.55 percentage points in normal times, while average wages rise by 0.057 and 0.049, respectively. This is not a contradiction: in the baseline, informal submarkets are attractive because they offer an accessible route out of unemployment, especially for low-skilled workers during downturns. When informality is shut down, workers lose this adjustment margin, but the remaining matches are entirely formal and are characterized by higher wages and lower separation risk. This reduction in job instability is strong enough to offset the loss of search access for low-skilled workers, raising their average welfare. For high-skilled workers, however, welfare falls: although they did not resort to the informal sector frequently in the baseline economy, the loss of this search margin and the resulting drop in continuation values outweigh the gains from higher average wages and lower unemployment. The aggregate comparison therefore reflects a fundamental equilibrium trade-off between search access and job stability, rather than a simple ranking of formal and informal jobs.

Table 9 decomposes the aggregate effects into the two margins emphasized by the model: access to jobs and job stability. Removing informality generally weakens search access. For all workers, equilibrium market tightness falls from 14.10 to 12.16 in recessions and from 23.99 to 21.67 in normal times, with corresponding declines in job-finding probabilities. The decline is especially pronounced for low-skilled workers in normal times, where market tightness falls by

5.55 and the job-finding probability declines by 1.55 percentage points.[6] At the same time, the no-informality economy features systematically lower separation rates across worker groups and aggregate states. For low-skilled workers, separations fall by 0.28 percentage points in recessions and 0.23 percentage points in normal times; for high-skilled workers, the corresponding declines are 0.18 and 0.13 percentage points. These patterns show that the counterfactual does not improve labor-market outcomes by making jobs easier to find. Instead, it replaces an accessible but less stable informal margin with a set of more stable formal matches.

Table 9: Search Access and Job Stability

		Recession			Normal		
Group	Metric	Baseline	No Inf.	Diff.	Baseline	No Inf.	Diff.
All workers	Market tightness	14.10	12.16	-1.94	23.99	21.67	-2.32
	Job-finding probability	9.64	9.24	-0.40	11.77	11.11	-0.66
	Separation rate	3.23	3.01	-0.22	1.01	0.84	-0.17
Low-skilled	Market tightness	8.94	9.09	0.15	17.66	12.10	-5.55
	Job-finding probability	8.29	8.35	0.06	10.65	9.10	-1.55
	Separation rate	2.89	2.60	-0.28	1.04	0.81	-0.23
High-skilled	Market tightness	17.61	14.14	-3.46	28.81	29.03	0.21
	Job-finding probability	10.55	9.81	-0.74	12.62	12.65	0.03
	Separation rate	3.46	3.27	-0.18	0.99	0.85	-0.13

Notes: The table reports average equilibrium job-finding probabilities and separation rates by worker group and period type, in percentage points, and average equilibrium market tightness in the submarkets of choice.

In this calibration, the stability margin is strong enough to reduce unemployment despite weaker search access, a pattern that is also visible in the spell evidence below. Figure 6 reports Kaplan–Meier survival curves for unemployment and job spells, pooled across human capital. The no-informality economy generates slightly more persistent unemployment spells, consistent with weaker search access, but also substantially more persistent job spells, consistent with lower separation risk.

This pattern is clarified in Table 10, which decomposes durations and churn by human capital level. For low-h workers (h_1), mean unemployment spells rise from 10.35 to 11.25 months, while mean job spells increase from 54.57 to 62.28 months and job spells per worker fall from 9.02 to 7.85. For middle-h workers (h_3), mean job spells rise from 46.97 to 57.34 months and churn decreases (job spells per worker fall from 10.46 to 8.82). For the highest grid point (h_5), the baseline and no-informality columns coincide: in the calibrated baseline these workers search in the formal sector in virtually all aggregate states (Figure 3), so shutting down informal vacancies leaves their reallocation patterns—and hence spell statistics—unchanged. Taken together, these spell statistics show the equilibrium trade-off in levels: removing informality weakens search access in the aggregate, but the remaining matches are sufficiently more stable

[6] Normal times include the median state z_3, in which many low-skilled workers choose to search in the informal sector in the baseline economy; see Figure 3.

to reduce unemployment.

Figure 6: Duration of Unemployment and Job Spells

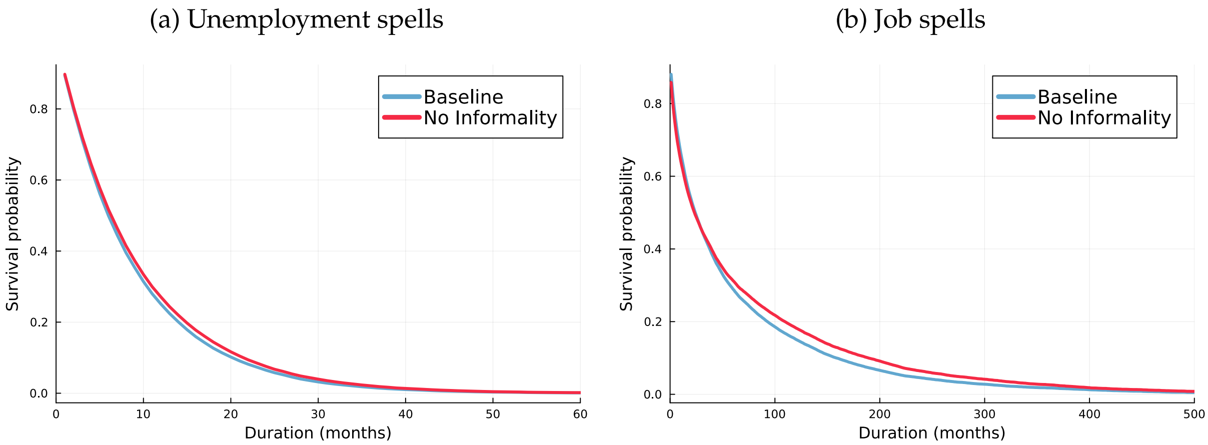


Table 10: Spell Durations and Churn by Human Capital

h	Baseline					No Informality				
	Unemployment		Job		N. jobs	Unemployment		Job		N. jobs
	Mean	Median	Mean	Median	per worker	Mean	Median	Mean	Median	per worker
h_1	10.35	8	54.57	31	9.02	11.25	8	62.28	33	7.85
h_2	10.11	7	50.39	28	9.85	11.15	8	57.08	27	8.58
h_3	8.51	6	46.97	23	10.46	8.62	6	57.34	26	8.82
h_4	8.13	6	46.99	23	10.57	8.83	6	48.54	18	10.03
h_5	8.53	6	48.47	18	10.10	8.53	6	48.47	18	10.10

Notes: The table reports mean and Kaplan–Meier median unemployment and job spell durations, in months, and the mean number of job spells per worker.

The empirical patterns in Section 2 emphasize both easier access to informal jobs when formal hiring weakens and much higher separation rates on informal ties. The counterfactual sharpens the model's account of that tension: here, "removing informality" is a structural shutdown of informal vacancies holding primitive shocks fixed, not a reduced-form experiment on stocks in the data. Under our preliminary calibration (Section 4), the dominant net effect on unemployment and on low-skilled welfare is driven by the stability margin—more durable formal matches—even though informal search improves exit from unemployment in the baseline. In addition, equilibrium wages in the simulated economy lie above the minimum wage in both sectors, so the minimum wage does not bind as a floor in the quantitative experiments reported here. That is a substantive restriction in real labor markets—where informality is sometimes linked to evasion of a binding formal-sector floor—and its irrelevance in our calibration may limit how much informal vacancies matter for reallocating labor in deep downturns relative to an environment in which the constraint bites. We interpret these results as disciplined qualitative lessons about how search access and job stability interact, rather than as final quantitative predictions for aggregate informality reform.

6 Concluding Remarks

This paper develops a quantitative search-and-matching framework with worker heterogeneity and an informal sector to study how informality shapes labor-market dynamics over the business cycle in a developing economy. The empirical evidence from Brazilian microdata shows that informality is closely linked to worker type and aggregate conditions: informal jobs provide an important employment margin, but they are also substantially more fragile than formal jobs. The model embeds these forces in a directed-search environment in which firms choose whether to create or destroy vacancies, workers choose where to search, and both decisions respond to aggregate productivity. Its block recursive structure is the key tractability device for combining worker heterogeneity and aggregate shocks into a search and matching model with informality.

The quantitative exercise highlights a central mechanism: informality affects labor-market outcomes through both search access and job stability. In the baseline economy, informal vacancies expand the set of submarkets available to unemployed workers, especially for low-skilled workers and in states in which formal hiring is weak. This margin helps workers exit unemployment, but it also exposes them to jobs with higher separation risk. The no-informality counterfactual makes this trade-off explicit. Removing informal vacancies weakens search access, but it shifts employment toward more stable formal matches. Under the current calibration, the stability margin is strong enough to reduce unemployment and raise average welfare for low-skilled workers, even though informal search is valuable in the baseline equilibrium. For high-skilled workers, the welfare effect is negative, reflecting changes in continuation values and the loss of an additional search margin.

These results should be interpreted with the appropriate discipline. The current calibration reproduces the main qualitative patterns emphasized in the data, including countercyclical unemployment and informality and greater exposure to informality among low-skilled workers. At the same time, it still faces quantitative tensions in matching the volatility of aggregate activity, unemployment, and transition rates. In addition, simulated wages lie above the minimum wage in the experiments reported here, so the model does not yet capture an environment in which the formal wage floor binds tightly and informality serves as a direct margin of regulatory evasion. The counterfactual results are therefore best read as disciplined qualitative lessons about how informality reshapes search access, match survival, and welfare, rather than as final quantitative predictions for aggregate informality reform.

The next steps follow directly from these diagnostics. A first priority is to tighten the mapping between empirical and simulated cyclical objects, including the measurement of aggregate states and finite-sample cyclical moments. Once these margins are disciplined, the framework can make two contributions that are difficult to obtain from the data alone. First, it can evaluate whether the same intervention has different effects depending on the phase of the business cycle, when workers' outside options, firms' vacancy incentives, and separation risks differ. Second, it can unbundle what is observed in the data as a single "formal job." Formal employment simultaneously reflects several regulatory features, including minimum-wage rules, income and payroll taxes, and separation costs. Because these institutions are active at the same time in the data, their separate contributions to labor-market dynamics and sectoral allocation are hard to identify empirically. A structural model provides a way to vary each margin separately and quantify which components of formal regulation matter most, for which workers, and at which points in the cycle.

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Appendix

A Existence and Uniqueness of the Block Recursive Equilibrium

This section provides the proof for the existence of a unique Block Recursive Equilibrium (BRE) for the model. The proof leverages the Banach Fixed-Point Theorem in a two-stage process, exploiting the decoupling between firm values/market tightness and worker values enabled by the absence of on-the-job search.

A.1 Model Setup Summary

We repeat here the main components of the model necessary for the proof. The states of the model are:

- Skill level $h \in \mathcal{H} = \{h_1, \ldots, h_{N_h}\}$, a finite set.
- Sector $s \in \mathcal{S} = \{F, I\}$.
- Fixed-wage contracts with $w \in [\underline{w}, \overline{w}]$.
- Idiosyncratic match productivity $y \in [\underline{y}, \overline{y}]$ drawn once from continuous CDF $G(y)$.
- Aggregate productivity $z \in Z = \{z_1, \ldots, z_{N_z}\}$, a finite set.

Let $x = (h, s, w, y; z)$ be the state of an existing match, $x' = (h, s, w, y; z')$ be the state tomorrow and $\eta = (h, s, w; z)$ be a submarket/vacancy type.

Value of an existing match to the firm Firms enjoy a flow profit

$$\pi(x) = \left(1 - \tau^p(s)\right)\left(e^{A_s + z + y} \times h - \left(1 + \tau^f(s)\right)w\right).$$

The firm chooses whether to fire or continue the match, accounting for exogenous separation cost $C(s)$:

$$J(x) = \pi(x) + \beta \mathbb{E}_{z'}\left[\max\left\{-C(s), (1 - \delta(s))J(x') - \delta(s)C(s)\right\}\middle| z\right] \tag{A1}$$

Given the fixed point J^* satisfying Eq. (A1), define the optimal total separation probability $d^*(x')$ for a match in state x':

$$d^*(x') = \begin{cases} 1 & \text{if } -C(s) > J^*(x') \\ \delta(s) & \text{otherwise.} \end{cases} \tag{A2}$$

This $d^*(x')$ is a well-defined function determined solely by J^*.

Value of posting a vacancy Let $V(\eta)$ be the expected net value of posting a vacancy in submarket η. The free entry condition requires $V(\eta) \leq 0$ for all η. In any active submarket where $\theta(\eta) > 0$, it must be that $V(\eta) = 0$. This implies:

$$\kappa_s = q\big(\theta(\eta)\big)\mathbb{E}_y\big[J(h, s, w, y; z)\big]$$

Solving for the equilibrium tightness $\theta^*(\eta)$ gives:

$$\theta^*(\eta) = \begin{cases} q^{-1}\left(\frac{\kappa_s}{\mathbb{E}_y[J^*(h,s,w,y;z)]}\right) & \text{if } \mathbb{E}_y\big[J^*(h, s, w, y; z)\big] > \kappa_s \\ 0 & \text{if } \mathbb{E}_y\big[J^*(h, s, w, y; z)\big] \leq \kappa_s, \end{cases} \tag{A3}$$

where J^* is the equilibrium value function.

Value of an unemployed worker

$$U(h, z) = b(h) + \beta\mathbb{E}_{z'}\big[S^U(h, z')\big|z\big], \tag{A4}$$

where $S^U(h, z)$ is the value of searching,

$$S^U(h, z) = U(h, z) + M(h, z), \tag{A5}$$

and $M(h, z)$ is the maximum expected gain from search,

$$M(h, z) = \max_{(s,w)} p\big(\theta^*(h, s, w; z)\big)\Big(\mathbb{E}_y\big[W(h, s, w, y, z)\big] - U(h, z)\Big). \tag{A6}$$

Value of an employed worker

$$W(x) = (1 - \tau^w(s))w + \beta\mathbb{E}_{z'}\Big[W(x') + d^*(x')\Big(B(s)w + S^U(h, z') - W(x')\Big)\Big|\, z\Big]. \tag{A7}$$

A.2 Assumptions

Assumption 1 (DRW Matching Function and Productivity Distribution)**.**

(i) The matching function takes the form:

$$M(u, v) = \frac{uv}{(u^\alpha + v^\alpha)^{1/\alpha}}$$

for some $\alpha > 0$, which implies:

$$p(\theta) = \frac{\theta}{(1 + \theta^\alpha)^{1/\alpha}}$$
$$q(\theta) = \frac{1}{(1 + \theta^\alpha)^{1/\alpha}}$$

(ii) The productivity distribution $G : [\underline{y}, \overline{y}] \to [0,1]$ is continuous with full support on $[\underline{y}, \overline{y}]$.

Assumption 2 (Parameter Restrictions)**.** *All tax rates $\tau^w, \tau^f, \tau^p \in [0,1)$, UI benefit rate $B \geq 0$, firing cost $C \geq 0$, and separation rates $\delta_F, \delta_I \in [0,1)$ are finite. The discount factor is $\beta \in (0,1)$. Home production $b(h)$ is bounded for all $h \in \mathcal{H}$.*

A.3 Proof of Existence and Uniqueness

The proof proceeds in two stages, using the Banach Fixed-Point Theorem.

A.3.1 Stage 1: Existence and Uniqueness of J^*

Proposition 1. *Under Assumptions 1 and 2, there exists a unique, bounded, and continuous firm value function $J^*(h, s, w, y, z)$ satisfying Eq. (A1).*

Proof. Define the operator T_J on the space $\mathcal{J}$ of bounded, continuous functions $J : \mathcal{H} \times \mathcal{S} \times [\underline{w}, \overline{w}] \times [\underline{y}, \overline{y}] \times Z \to \mathbb{R}$.

$$(T_J J_{old})(x) = \pi(x) + \beta \mathbb{E}_{z'} \left[\max\{-C(s), (1-\delta(s)) J_{old}(x') - \delta(s) C(s)\} \middle| z \right]$$

1. **State Space Completeness:** The state space is $\mathcal{X} = \mathcal{H} \times \mathcal{S} \times [\underline{w}, \overline{w}] \times [\underline{y}, \overline{y}] \times Z$. Since $\mathcal{H} = \{h_1, \ldots, h_{N_h}\}$ and $\mathcal{S} = \{F, I\}$ are finite, $[\underline{w}, \overline{w}]$ and $[\underline{y}, \overline{y}]$ are compact intervals, and Z is finite, $\mathcal{X}$ is compact. The space $\mathcal{J}$ of bounded continuous functions on $\mathcal{X}$ equipped with the supremum norm $\|J\| = \sup_{x \in \mathcal{X}} |J(x)|$ is thus a complete metric space.

2. **T_J maps $\mathcal{J}$ to $\mathcal{J}$:**

   - *Boundedness:* $\pi(x)$ is bounded on the compact domain $\mathcal{X}$. $C(s)$ is bounded by Assumption 2. If J_{old} is bounded, the term inside the max is bounded, and the expectation (a finite sum) is bounded. Thus $T_J J_{old}$ is bounded.
   - *Continuity:* $\pi(x)$ is continuous in all arguments. J_{old} is continuous by assumption. The term $(1-\delta(s))J_{old}(x') - \delta(s)C(s)$ is continuous in all state variables. The $\max\{A, B\}$ of two continuous functions is continuous. The expectation $\mathbb{E}_{z'}[\cdot] = \sum_{z' \in Z} \text{prob}(z'|z) \cdot (\cdot)$ is a finite sum of continuous functions weighted by transition probabilities, hence continuous in the current state variables (h, s, w, y, z). Therefore $T_J J_{old}$ is continuous.

3. **T_J is a Contraction:** We verify Blackwell's sufficient conditions.

   - *Monotonicity:* If $J_A \geq J_B$ pointwise, then

   $$(1-\delta(s))J_A(x') - \delta(s)C \geq (1-\delta(s))J_B(x') - \delta(s)C$$

   for all x'. The max operator preserves monotonicity. Since $\beta \geq 0$ and transition probabilities are non-negative, $\mathbb{E}_{z'}[\cdot]$ preserves monotonicity. Thus $T_J(J_A) \geq T_J(J_B)$.

- *Discounting:* For $a > 0$ and any function J,

$$\begin{aligned}
T_J(J + a) &= \pi(x) + \beta\mathbb{E}_{z'}\Big[\max\big\{-C(s), (1-\delta(s))(J(x') + a) - \delta(s)C(s)\big\}\Big| z\Big] \\
&= \pi(x) + \beta\mathbb{E}_{z'}\Big[\max\big\{-C(s), (1-\delta(s))J(x') - \delta(s)C(s) + (1-\delta(s))a\big\}\Big| z\Big] \\
&\le \pi(x) + \beta\mathbb{E}_{z'}\Big[\max\big\{-C(s), (1-\delta(s))J(x') - \delta(s)C(s)\big\} + (1-\delta(s))a\Big| z\Big] \\
&= T_J(J) + \beta(1-\delta(s))a
\end{aligned}$$

  where the inequality follows from the general property that $\max\{A, B + c\} \le \max\{A, B\} + c$ for $c \ge 0$.
  Let $\underline{\delta} = \min_{s\in\mathcal{S}}\{\delta(s)\} \ge 0$. Then $(1 - \delta(s)) \le (1 - \underline{\delta})$ for all s, so:

$$T_J(J + a) \le T_J(J) + \beta(1 - \underline{\delta})a$$

  The modulus of contraction is $\gamma_J = \beta(1 - \underline{\delta})$. Since $\beta \in (0, 1)$ (Assumption 2) and $\underline{\delta} \in [0, 1)$, we have $\gamma_J < 1$.

4. **Conclusion:** By Blackwell's Sufficient Conditions, T_J is a contraction mapping. By the Banach Fixed-Point Theorem, T_J has a unique fixed point $J^* \in \mathcal{J}$, which is bounded and continuous.

□

Corollary 1 (Continuity of θ^* with DRW Matching)**.** *Assume the matching function takes the form $M(u, v) = \frac{uv}{(u^\alpha + v^\alpha)^{1/\alpha}}$ for $\alpha > 0$. Then the equilibrium market tightness function $\theta^*(\eta)$ defined by Eq. (A3) is continuous in $\eta = (h, s, w; z)$.*

Proof. With the DRW matching function, we have:

$$q(\theta) = \frac{1}{(1 + \theta^\alpha)^{1/\alpha}}$$

which is continuous and strictly decreasing with $q(0) = 1$ and $\lim_{\theta\to\infty} q(\theta) = 0$.

Continuity of the expected firm value: We first establish that $\mathbb{E}_y\left[J^*(h, s, w, y; z)\right]$ is continuous in $\eta = (h, s, w; z)$. Let $\{\eta_n\}$ be any sequence with $\eta_n \to \eta$ and write $J^*(\eta_n, y) = J^*(h_n, s_n, w_n, y; z_n)$. We need to show:

$$\lim_{n\to\infty} \mathbb{E}_y\left[J^*(\eta_n, y)\right] = \mathbb{E}_y\left[J^*(\eta, y)\right]$$

.

We apply the Dominated Convergence Theorem, which requires three conditions:

1. **Pointwise convergence:** For each fixed $y \in [\underline{y}, \overline{y}]$, we have $J^*(\eta_n, y) \to J^*(\eta, y)$ as $n \to \infty$.

Justification: Since J^* is continuous in all its arguments (Proposition 1), we have

$$J^*(h_n, s_n, w_n, y; z_n) \longrightarrow J^*(h, s, w, y; z)$$

for any $y \in [\underline{y}, \overline{y}]$.

2. **Integrable dominating function:** There exists an integrable function $g(y)$ such that $|J^*(\eta_n, y)| \le g(y)$ for all n and all $y \in [\underline{y}, \overline{y}]$.

   Justification: From Proposition 1, the function J^* is bounded on the compact state space. Let $M = \sup_x |J^*(x)| < \infty$. Then:

   $$|J^*(\eta_n, y)| \le M \text{ for all } n, y$$

   We can take $g(y) = M$, which is integrable: $\int_{\underline{y}}^{\overline{y}} g(y)\, dG(y) = M < \infty$.

3. **Integration with respect to a probability measure:** The distribution G defines a probability measure on $[\underline{y}, \overline{y}]$.

   This is satisfied by construction.

By the Dominated Convergence Theorem:

$$\begin{aligned}
\lim_{n\to\infty} \mathbb{E}_y[J^*(\eta_n, y)] &= \lim_{n\to\infty} \int_{\underline{y}}^{\overline{y}} J^*(\eta_n, y)\, dG(y) \\
&= \int_{\underline{y}}^{\overline{y}} \lim_{n\to\infty} J^*(\eta_n, y)\, dG(y) \quad \text{(interchanging limit and integral)} \\
&= \int_{\underline{y}}^{\overline{y}} J^*(\eta, y)\, dG(y) \\
&= \mathbb{E}_y[J^*(\eta, y)]
\end{aligned}$$

Since this holds for any sequence $\eta_n \longrightarrow \eta$, the function $\mathbb{E}_y[J^*(\eta, y)]$ is continuous in η by the sequential characterization of continuity.

Interior of active region: Where $\mathbb{E}_y[J^*(\eta, y)] > \kappa_s$, the inverse function

$$q^{-1}(q) = \left(q^{-\alpha} - 1\right)^{1/\alpha}$$

is continuous (as a composition of continuous functions: power, subtraction, and root). Therefore, $\theta^*(\eta) = q^{-1}\left(\frac{\kappa_s}{\mathbb{E}_y[J^*(\eta, y)]}\right)$ is continuous as a composition of:

- The continuous function $\eta \mapsto \mathbb{E}_y[J^*(\eta, y)]$ (established above).
- The continuous function $r \mapsto \kappa_s / r$ (on $r > \kappa_s$).
- The continuous function q^{-1}.

At the boundary: Where $\mathbb{E}_y[J^*(\eta, y)] = \kappa_s$, as $\eta \to \eta_0$ with $\mathbb{E}_y[J^*(\eta, y)] \to \kappa_s^+$, we have:

$$\frac{\kappa_s}{\mathbb{E}_y[J^*(\eta, y)]} \to \frac{\kappa_s}{\kappa_s} = 1$$

.

By continuity of q^{-1} and the explicit formula:

$$\lim_{\eta \to \eta_0} \theta^*(\eta) = \lim_{q \to 1^-} q^{-1}(q) = \lim_{q \to 1^-} \left(q^{-\alpha} - 1\right)^{1/\alpha} = (1^{-\alpha} - 1)^{1/\alpha} = 0^{1/\alpha} = 0 = \theta^*(\eta_0)$$

Therefore, $\theta^*(\eta)$ is continuous everywhere. □

A.3.2 Stage 2: Existence and Uniqueness of (U^*, W^*)

Given the unique J^*, we derive V^*, θ^*, and the total separation probability function $d^*(x')$ (from Eq. A2) as fixed, bounded functions determined by J^*. Since J^* is continuous (Proposition 1), the function $d^*(x')$ is Borel measurable (continuous functions are measurable, and indicator functions of measurable sets are measurable). Furthermore, θ^* is continuous (Corollary 1).

Proposition 2. *Under Assumptions 1 and 2, given J^*, there exists a unique pair of bounded and measurable worker value functions (U^*, W^*) satisfying Eqs. (A4), (A5), (A7).*

Proof. Define the operator T_W on the space $\mathcal{W}$ of pairs of bounded, Borel measurable functions $\mathbf{V}_{old} = (U_{old}, W_{old})$ defined on their respective state spaces. The space $\mathcal{W}$ is equipped with the norm $\|\mathbf{V}\| = \max(\|U\|_\infty, \|W\|_\infty)$ where $\|\cdot\|_\infty$ denotes the supremum norm.

$$T_W(\mathbf{V}_{old}) = (U_{new}, W_{new}),$$

where

$$\begin{aligned}
W_{new}(x) &= \left(1 - \tau^w(s)\right)w + \beta \mathbb{E}_{z'}\left[(1 - d^*(x'))W_{old}(x') + d^*(x')\left(B(s)w + S^U_{old}(h, z')\right)\middle| z\right] \\
U_{new}(h, z) &= b(h) + \beta \mathbb{E}_{z'}\left[S^U_{old}(h, z')\middle| z\right],
\end{aligned}$$

and $S^U_{old}(h, z) = U_{old}(h, z) + M_{old}(h, z)$, with M_{old} defined by Eq. (A6) using $\mathbf{V}_{old}$.

1. **State Space Completeness:** The state spaces for U and W are compact (finite sets for discrete components, compact intervals for continuous components). The space $\mathcal{W}$ of bounded, Borel measurable functions equipped with the supremum norm is thus a complete metric space.

2. **T_W maps $\mathcal{W}$ to $\mathcal{W}$:**

   - *Boundedness:*
     - For W_{new}: The flow wage $(1 - \tau^w(s))w$ is bounded on the compact domain. The expectations involve bounded functions W_{old}, U_{old} (by assumption), bounded $d^* \in [\underline{\delta}, 1]$, bounded $B(s)$, and bounded w. Therefore W_{new} is bounded.

- For U_{new}: The flow utility $b(h)$ is bounded by Assumption 2. The expectation of the bounded function S^U_{old} is bounded. Therefore U_{new} is bounded.

- *Measurability:*
  - The objective function in $M_{old}(h,z)$ is measurable in (s,w) because it depends on measurable functions W_{old}, U_{old}, the continuous CDF G, the continuous function θ^* (Corollary 1), and the continuous function $p(\cdot)$. The maximum over a compact set of a measurable function yields a measurable function (Measurable Maximum Theorem). Thus $M_{old}(h,z)$ is measurable, and hence S^U_{old} is measurable.
  - W_{new} is measurable as a composition and expectation of measurable functions: W_{old} and S^U_{old} are measurable by assumption, d^* is measurable, flow wages are continuous (hence measurable), and expectations preserve measurability.
  - U_{new} is measurable as an expectation of the measurable function S^U_{old}.

3. T_W **is a Contraction:** We show $\|T_W(\mathbf{V}_A) - T_W(\mathbf{V}_B)\| \leq \beta\|\mathbf{V}_A - \mathbf{V}_B\|$.

- Let $\|\mathbf{V}_A - \mathbf{V}_B\| = d = \max(\|W_A - W_B\|_\infty, \|U_A - U_B\|_\infty)$.
- For any state (h,z), let $m_A = (s_A, w_A)$ be the optimal choice maximizing $M_A \equiv M(h,z;\mathbf{V}_A)$ and m_B be optimal for $\mathbf{V}_B$.
- **Step 1: Bound on $|S^U_A - S^U_B|$:**
  By optimality of m_A for $\mathbf{V}_A$:

$$M_A \geq p(\theta^*(h,m_B;z))\Big(\mathbb{E}_y\big[W_A(h,m_B,y,z)\big] - U_A(h,z)\Big)$$

Therefore:

$$\begin{aligned}
S^U_A - S^U_B &= (U_A - U_B) + (M_A - M_B) \\
&\geq (U_A - U_B) + p_B\Big(\mathbb{E}_y\big[W_A(\cdot)\big] - U_A\Big) - p_B\Big(\mathbb{E}_y\big[W_B(\cdot)\big] - U_B\Big) \\
&= \big(U_A - U_B\big)\big(1 - p_B\big) \;+\; p_B\,\mathbb{E}_y\Big[W_A(\cdot) - W_B(\cdot)\Big],
\end{aligned}$$

where $p_B = p(\theta^*(h,m_B;z))$.
Since $U_A - U_B \geq -d$ and $W_A - W_B \geq -d$ pointwise, and $p_B \in [0,1]$:

$$S^U_A - S^U_B \geq (-d)(1-p_B) + p_B(-d) = -d$$

By symmetry (starting from $S^U_B - S^U_A$ and using optimality of m_B):

$$S^U_B - S^U_A \geq -d$$

Therefore $|S^U_A - S^U_B| \leq d$ for all (h,z), which implies $\|S^U_A - S^U_B\|_\infty \leq d$.

- **Step 2: Contraction for U:**

$$\begin{aligned}\left|U_A^{new}(h,z) - U_B^{new}(h,z)\right| &= \left| \beta \mathbb{E}_{z'}\left[S_A^U(h,z') - S_B^U(h,z') \mid z\right] \right| \\ &\le \beta \mathbb{E}_{z'}\left[\left|S_A^U(h,z') - S_B^U(h,z')\right| \mid z\right] \\ &\le \beta \mathbb{E}_{z'}\left[d \mid z\right] = \beta d\end{aligned}$$

- **Step 3: Contraction for W:**

$$\begin{aligned}&\left|W_A^{new}(x) \;-\; W_B^{new}(x)\right| \\ &\quad= \left| \beta \mathbb{E}_{z'}\left[(1-d^*(x'))\big(W_A(x') - W_B(x')\big) + d^*(x')\big(S_A^U(h,z') - S_B^U(h,z')\big) \mid z\right] \right| \\ &\quad\le \beta \mathbb{E}_{z'}\left[\, (1-d^*(x'))\left|W_A(x') - W_B(x')\right| \;+\; d^*(x')\left|S_A^U(h,z') - S_B^U(h,z')\right| \mid z \,\right] \\ &\quad\le \beta \mathbb{E}_{z'}\left[\, (1-d^*(x'))d + d^*(x')d \mid z \,\right] \\ &\quad= \beta d\end{aligned}$$

- Therefore:

$$\|T_W(\mathbf{V}_A) - T_W(\mathbf{V}_B)\| = \max(\|W_A^{new} - W_B^{new}\|_\infty, \|U_A^{new} - U_B^{new}\|_\infty) \le \beta d,$$

so T_W is a contraction with modulus $\beta < 1$.

4. **Conclusion:** By the Banach Fixed-Point Theorem, T_W has a unique fixed point $\mathbf{V}^* = (U^*, W^*) \in \mathcal{W}$, which is bounded and measurable.

□

A.4 Conclusion

The two-stage proof demonstrates the existence and uniqueness of the firm value function J^* (bounded and continuous) and the worker value functions (U^*, W^*) (bounded and measurable). These value functions, along with the derived firm vacancy value V^*, market tightness θ^* (continuous), and associated optimal policy functions (worker search choice $m^*(h,z)$ and firm firing rule defining d^*), constitute the unique Block Recursive Equilibrium for the specified economy.

Note on Block Recursivity

The equilibrium derived in this proof possesses the Block Recursive property. This is because the core equilibrium objects determining market interactions – namely the firm's value function for existing matches (J^*), the firm's expected value from posting a vacancy (V^*), the resulting market tightness (θ^*), and the firm's optimal firing rule (d^*) – are all determined in Stage 1 of the proof *independently* of the worker value functions (U^*, W^*) and, crucially, independently of the aggregate distribution of unemployed workers.

This decoupling occurs because the model assumes no on-the-job search, preventing the distribution of employed workers from affecting market tightness. Consequently, the applicant pool for any vacancy consists only of unemployed workers of a specific type. As search is directed, everyone looking for a job in a particular submarket is willing to accept the job if matched to a firm in that submarket.

Stage 2 then solves for the worker value functions taking the results from Stage 1 as given parameters. This structure allows the "block" of value functions and prices to be solved separately from the aggregate state distribution, fulfilling the definition of a BRE. This contrasts with models featuring on-the-job search, where the interdependence between firm values, worker search strategies, and the worker distribution typically necessitates different proof techniques.

B Latent Human Capital and Empirical Skill Shares

This section documents the numerical methods used to discretize a continuous latent human-capital distribution and to map latent productivity states to observed skill labels in the data. The model state is latent human capital h, while the data are reported for labeled binary groups $j \in \{L, H\}$. The link between the two is a set of Bayesian weights $\omega_{H,k} \equiv P(H \mid h_k)$ defined below.

B.1 Demographic Mapping

Distributional Assumptions We assume latent human capital h follows a Log-Normal distribution for each demographic group $j \in \{L, H\}$:

$$\ln(h) \mid j \sim \mathcal{N}(\mu_j, \sigma_h^2)$$

The structural parameters governing these distributions are the common dispersion parameter (σ_h) and the two location parameters (μ_L and μ_H).

The Mixture Distribution Let π_H be the fixed empirical share of high-skilled workers (calculated from the PME sample average). The aggregate economy-wide Probability Density Function (PDF) and Cumulative Distribution Function (CDF) are:

$$f_{mix}(h) = (1 - \pi_H) f(h \mid \mu_L, \sigma_h) + \pi_H f(h \mid \mu_H, \sigma_h)$$
$$F_{mix}(h) = (1 - \pi_H) F(h \mid \mu_L, \sigma_h) + \pi_H F(h \mid \mu_H, \sigma_h)$$

where f and F are the standard Log-Normal PDF and CDF, respectively.

Bayesian Weight Calculation Since the model uses productivity states (h) but the data uses demographic labels (L, H), we link them via conditional probabilities. For each grid point h_k, define the posterior probability that a worker with latent productivity h_k belongs to the high-skill group:

$$\omega_{H,k} \equiv P(H \mid h_k) = \frac{\pi_H f(h_k \mid \mu_H, \sigma_h)}{(1 - \pi_H) f(h_k \mid \mu_L, \sigma_h) + \pi_H f(h_k \mid \mu_H, \sigma_h)}. \tag{B1}$$

The complementary weight is $\omega_{L,k} = 1 - \omega_{H,k}$. These weights are pre-calculated once per estimation step and stored. They are used exclusively for moment aggregation, not for agent decision-making.

Normalization of the Human Capital Scale The latent human-capital state h is measured in arbitrary units. Because the model is homogeneous in productivity scales, only *relative* differences in h are economically meaningful. To pin down the unit of measurement and avoid scale indeterminacy, we normalize the mean human capital of the low-skill labeled group to one:

$$\mathbb{E}[h \mid L] = 1.$$

Under the log-normal specification assumed above, this normalization implies $\mathbb{E}[h \mid L] = \exp(\mu_L + \sigma_h^2/2) = 1$, and therefore $\mu_L = -\frac{\sigma_h^2}{2}$. The other two parameters (μ_H and σ_h) are estimated via SMM (see Appendix C for details).

This normalization has two practical advantages. First, it makes the high-skill location parameter μ_H interpretable as a *relative shift* of the H distribution with respect to the normalized L distribution, which is naturally tied to skill-premium moments. Second, with home production parameterized as $b(h) = \rho_b h$, the parameter ρ_b can be interpreted as a replacement-rate slope in the same units as productivity and wages, helping connect unemployment-related moments to the model's outside option. The population share parameter π_H governs mixture composition and does not affect this scale normalization.

B.2 Grid Construction

The model assumes a continuos latent human capital variable h. To solve the model numerically, we need to discretize it into a grid $\mathcal{H} = \{h_1, \ldots, h_{N_h}\}$. This grid is constructed to efficiently capture the probability mass of the population using a Quantile-Mixture method. We use N_h grid points and set them such that each point represents an equal mass $1/N_h$ of the aggregate population. The algorithm is as follows:

Step 1: Define Target Quantiles. Calculate the cumulative probability targets for the midpoint of N_h equal-mass bins:

$$p_k = \frac{k - 0.5}{N_h}, \quad \text{for } k = 1, \ldots, N_h$$

Step 2: Numerical Inversion. Solve for the productivity levels h_k corresponding to these quantiles. Since F_{mix} has no closed-form inverse, use a numerical root-finder to find the unique root h_k for each p_k:

$$F_{mix}(h_k) - p_k = 0.$$

Output: a sorted vector of grid points $\mathcal{H} = [h_1, h_2, \ldots, h_{N_h}]$.

B.3 Simulation and Moment Aggregation

Initialization In our simulation exercises, we build a panel of N agents.

- Each agent n is assigned a permanent productivity index $k_n \in \{1, \ldots, N_h\}$.
- Because the grid was constructed using equal-mass quantiles, we draw k_n from a discrete uniform distribution:

$$k_n \sim \text{Uniform}\{1, N_h\}$$

This simple sampling correctly reproduces the complex aggregate mixture density $f_{mix}(h)$ in the cross-section. In the SMM estimation, the index draws $\{k_n\}_{n=1}^{N}$ are generated once and held fixed across iterations; only the grid values $\{h_k\}$ change with (μ_H, σ_h).

Calculating Weighted Moments To match the model outputs to the PME data moments (which are split by demographic group), we use the Bayesian weights $\omega_{H,k}$ defined in Equation B1 to aggregate the simulated outcomes accordingly when constructing model moments by observed skill group.

C Data and Estimation Methodology

This section details how we use the data to discipline the model's parameters. It describes the externally calibrated parameters, the internally estimated parameters, the construction of the empirical target moments, and the Simulated Method of Moments (SMM) estimation procedure.

C.1 List of Parameters

The model contains a total of 23 structural parameters. We divide these into two groups: (1) externally calibrated parameters, and (2) internally estimated parameters (via SMM).

The set of externally calibrated parameters has their values drawn from the literature, from institutional rules, or calculated directly from the data. This approach reduces the dimensionality of the SMM estimation and anchors the model to key real-world observations. The list of externally calibrated parameters is presented in Table 5, in the main text.

The core of the estimation is a vector P of 14 structural parameters, which are jointly estimated in the SMM procedure. These parameters and their primary identifying moments are listed in Table C1.

Table C1: Internally Estimated Parameters

Symbol	Description	Primary Identifying Moment(s)
Long-run average moments		
μ_H	Relative human capital location	High-skilled wage premium
σ_h	Human capital dispersion	Overall wage dispersion
ρ_b	Home production replacement rate	Average UR and IR (by skill level)
A_F, κ_F, κ_I	Productivity & Vacancy costs	Informality rates and job-finding rates, and the formal-informal wage premium
δ_F, δ_I	Separation rates	Average levels of job-separation rates
$\bar{m}_w$	Minimum wage	Minimum wage ratio
μ_y, σ_y	Idiosyncratic shock parameters	Jointly identified with other parameters by the labor market stock and flow moments
Cyclical moments		
C	Firing costs	Cyclical sensitivity of job-separation rates
$\rho_z, \sigma_\varepsilon$	Aggregate shock parameters	Aggregate activity moments

C.1.1 Defining skill and skill-shares

In the data, we define high-skilled workers as those with at least a high school degree, and low-skilled workers as those with less than a high school degree. This binary classification aligns with the educational information available in the PME microdata and is the basis for the empirical moments reported by skill group. See Appendix B for details on how we link this observed classification to the model's latent human capital distribution.

To set the fixed skill shares π_H and $\pi_L = 1 - \pi_H$, we first calculate the share of high-skilled workers in the labor force for each month, creating a time series $\pi_H(t)$. We then set π_H as the average of this series over the full sample period.

C.1.2 Policy Parameters

Tax rates reflect Brazilian institutional features. The income tax rate is set at 11% income based on maximum effective tax rates from Sindifisco Nacional (2023). Payroll taxes have a 37.5% rate following the calculations from Ulyssea (2018) and Bosch and Esteban-Pretel (2012), which combine direct taxes and social security contributions. The profit tax rate τ^p is set to zero as a simplification.

Unemployment insurance payments depend on the worker's past wage, how long they were employed before layoff, and the interval since last receiving UI. The system provides 3 to 5 months of benefits, and there are floors (minimum wage) and ceilings to the amount of the benefits. To simplify, we assume $B = 3$, the minimum duration of benefits, as the payment in the model implies receiving the full wage (and not just a fraction of it).

C.2 Empirical Target Moments

We construct a vector M_{data} of 30 target moments from the data. The calculation of these moments requires a multi-step preparation process to ensure consistency between the empirical data and the model's simplified structure.

C.2.1 The Vector of Target Moments

Table C2: Empirical Target Moments

Labor Market Stocks (Avg. of education-detrended series)		
1	Avg. unemployment rate, low-skilled (%)	11.6671
2	Avg. unemployment rate, high-skilled (%)	11.4529
3	Avg. informality rate, low-skilled (%)	33.8790
4	Avg. informality rate, high-skilled (%)	19.0439
Labor Market Flows (Avg. of education-detrended and adjusted series)		
5	Avg. formal job-finding rate (UF), low-skilled (%)	8.2200
6	Avg. formal job-finding rate (UF), high-skilled (%)	9.4375
7	Avg. informal job-finding rate (UI), low-skilled (%)	12.6513
8	Avg. informal job-finding rate (UI), high-skilled (%)	8.0309
9	Avg. formal job-separation rate (FU), low-skilled (%)	0.8840
10	Avg. formal job-separation rate (FU), high-skilled (%)	0.9511
11	Avg. informal job-separation rate (IU), low-skilled (%)	3.0896
12	Avg. informal job-separation rate (IU), high-skilled (%)	3.2004
Wage Structure		
13	Formal-informal wage premium (log points)	0.0730
14	High-skilled vs. low-skilled premium (log points)	0.5255
15	Minimum wage ratio	0.6110
16	Std. dev. of log hourly wages, all employed workers	0.6730
Aggregate Activity Moments		
17	Std. dev. of cycle component of activity (log points)	0.0199
18	Lag-1 autocorrelation of cycle component of activity	0.9564
Labor Market Dispersion		
19	Std. dev. of unemployment rate, overall (%)	0.8972
20	Std. dev. of informality rate, overall (%)	0.6115
21	Std. dev. of formal job-finding rate (UF), overall (%)	0.9470
22	Std. dev. of informal job-finding rate (UI), overall (%)	0.8636
23	Std. dev. of formal job-separation rate (FU), overall (%)	0.1029
24	Std. dev. of informal job-separation rate (IU), overall (%)	0.4358
Labor Market Co-movement with Activity		
25	Correlation of unemployment rate and cycle activity	-0.5755
26	Correlation of informality rate and cycle activity	-0.3467
27	Correlation of formal job-finding (UF) and cycle activity	0.5580
28	Correlation of informal job-finding (UI) and cycle activity	0.0478
29	Correlation of formal job-separation (FU) and cycle activity	-0.3326
30	Correlation of informal job-separation (IU) and cycle activity	-0.3121

C.2.2 Data Preparation: Adjusting for Trends and Model Scope

The empirical labor market time series are affected by trends and data-model mismatches that must be accounted for.

1. Adjusting for Model Scope (Flows Adjustment) Our model is a closed 3-state system (Formal, Informal, Unemployed), but the data contains additional "exit" states corresponding to other employment conditions (self-employed, employers, public sector, and out of the labor force). To make the empirical data consistent with the model's structure, we must calculate flow probabilities conditional on remaining within the {F, I, U} system.

For any transition from an origin state $J \in \{F, I, U\}$ to a destination state $K \in \{F, I, U\}$, the adjusted probability is calculated as:

$$p_{JK}^{adj}(t) = \frac{\mathrm{N}_{J\to K}(t)}{N_J(t) - N_{J\to X}(t)}, \tag{C1}$$

where $N_J(t)$ is the total stock in state J at time t, and $N_{J\to X}(t)$ is the number of individuals flowing from state J to any "exit" state $X \notin \{F, I, U\}$. All flow rates time series (Moments 5-12) are constructed using this adjustment method to the raw series.

Stocks (unemployment and informality rates) are calculated directly from the raw data, also considering the appropriate denominators (only private sector workers and unemployed).

2. Seasonal Adjustment First, for all raw monthly aggregated time series (which include the adjusted flow series from the step above), we apply a standard X-13 ARIMA-SEATS procedure to remove predictable seasonal fluctuations. We then apply a centered moving average to smooth high-frequency noise.

3. Educational Trend Removal Second, we observe a significant, slow-moving trend in the educational attainment of the workforce over our sample period. Our model, by design, assumes a fixed skill share (π_H). To make the data consistent with the model's structure, we must remove the effect of this compositional trend. We do this by:

1. Extracting the low-frequency trend from the high-skill share series, $\pi_H^{trend}(t)$, using an HP filter.
2. For each seasonally-adjusted labor market series $Y(t)$ of the previous step, we run the regression $Y(t) = \alpha + \beta\pi_H^{trend}(t) + \epsilon(t)$.
3. We construct the detrended series as $Y_{detrended}(t) = \text{mean}(Y) + \epsilon(t)$. This new series will share the same average as the original series, so this step does not affect the long-run average moments. It is critical, however, for the calculation of the cyclical moments, by removing the demographic trend component.

4. Cyclical Trend Removal Finally, to calculate the cyclical moments, we isolate the high-frequency business cycle component applying the Hodrick-Prescott (HP) filter to the fully cleaned $Y_{detrended}(t)$ series from the previous step. This step is also applied to the activity measure (smoothed log of seasonally adjusted IBC-Br), yielding its cyclical component.

C.2.3 Wage Premiums

The two wage premium moments are derived from Mincer regressions on the microdata panel, as described in the main text. To maintain consistency with the model scope, we include only private formal and informal wage workers. The results of these two regressions, from which Moments 13 and 14 are taken, are presented in Table C3.

Table C3: Wage Premium Regressions

	Formal Premium	Skill Premium
Formal sector	0.073***	0.166***
	(0.001)	(0.001)
Age	0.041***	0.057***
	(0.001)	(0.000)
Age squared	-0.000***	-0.001***
	(0.000)	(0.000)
Female		-0.176***
		(0.001)
(Intercept)		0.650***
		(0.005)
High-skilled		0.525***
		(0.001)
Individual fixed effects	Yes	–
Industry dummies	Yes	Yes
Region dummies	–	Yes
N. Obs.	4,194,936	4,437,982
R^2	0.900	0.333
R^2 Adj.	0.873	0.333

Notes: * p < 0.1, ** p < 0.05, *** p < 0.01.

Table C4 shows substantial sorting across industries by both formality status and skill, supporting the inclusion of industry fixed effects in the empirical wage regressions. In particular, there are especially low formal and high-skill shares in domestic services and much higher shares in finance, education and health, and business services.

Table C4: Formal and High-Skill Shares by Industry

Industry	Formal share (%)	High-skill share (%)
Agro and extractive	73.1	46.1
Manufacturing	84.2	53.7
Utilities	91.7	55.3
Construction	65.9	30.3
Trade	77.2	59.4
Accommodation and food	71.3	40.2
Transport and communications	84.1	56.1
Finance	87.4	90.9
Business services and administration	84.1	63.2
Education and health	81.8	82.9
Other services	56.2	60.7
Domestic services	39.4	16.8
Government and residual	55.4	51.9

Notes: Shares computed among private formal and private informal wage workers with non-missing log hourly wages and positive hours.

C.2.4 Minimum wage ratio

The minimum wage ratio moment (Moment 15) is calculated as the ratio between the average real monthly statutory minimum wage over the sample period (set by law) and the average real monthly wage of low-skilled, full-time, private sector workers.

C.3 SMM Procedure

The SMM procedure consists in estimating the parameter vector P of Table C1 in a single, unified step by minimizing the distance between the data moments M_{data} (from Table C2) and the model-generated moments $M_{model}(P)$.

C.3.1 Weighting Matrix

To account for differences in the scale of the moments (e.g., rates vs. standard deviations) and to construct an efficient estimator, we use a weighting matrix. We estimate the variance-covariance matrix of the empirical microdata moments via a **block bootstrap** procedure with $B = 5000$ repetitions:

1. We resample with replacement from the unique individual identifiers in our microdata panel.
2. For each bootstrap replication, we build a new panel dataset.

3. We re-run the entire empirical pipeline on this new panel: calculating the adjusted flow rates and stocks, seasonal adjustment, educational detrending, HP-filtering, and the calculation of the full 30-moment vector.
4. We compute the covariance matrix $\hat{\Sigma}$ of the resulting $B \times 30$ matrix of moments for the microdata-based blocks.

Variance of Aggregate Activity Moments The block bootstrap described above resamples individual identifiers from the microdata panel and therefore captures the sampling uncertainty of moments that depend on the survey observations. However, the aggregate activity moments – the standard deviation and the lag-1 autocorrelation of the cycle component of activity – are computed from the IBC-Br time series and are invariant to the individual-level resampling. Their sampling uncertainty must therefore be estimated separately from the microdata bootstrap.

Our empirical activity series is constructed in several steps. We start from the seasonally adjusted IBC-Br index, take logs, smooth the logged series with a centered three-month moving average, extract the cyclical component with the HP filter on the full available macro sample, and only then restrict the resulting cycle to the PME sample horizon. Let $\{c_t\}_{t=1}^T$ denote this final filtered cycle over the PME window. The two target moments of the aggregate activity block are

$$m = \left(\hat{\sigma}_c, \hat{\rho}_c(1)\right)',$$

where $\hat{\sigma}_c$ is the sample standard deviation of $\{c_t\}_{t=1}^T$ and $\hat{\rho}_c(1)$ is its sample lag-1 autocorrelation.

Rather than imposing closed-form asymptotic formulas for the variance of these moments, we estimate their covariance matrix with a sieve bootstrap. The filtered cycle is approximately covariance-stationary in the PME sample and is well described by a finite-order autoregression. But the weight matrix requires the sampling uncertainty of the empirical moments themselves, not merely the standard errors of reduced-form time-series parameters. The sieve bootstrap is well suited to this setting because it preserves the dependence structure of the filtered cycle while delivering a direct estimate of the finite-sample distribution of $\hat{\sigma}_c$ and $\hat{\rho}_c(1)$.

Concretely, we approximate the filtered cycle by an autoregression of order p,

$$c_t = \alpha + \sum_{j=1}^{p} \phi_j c_{t-j} + u_t, \tag{C2}$$

where the lag order is selected from the empirical cycle and then held fixed throughout the bootstrap procedure. The empirical PME activity cycle is well approximated by an AR(8), so the sieve bootstrap is based on this finite-order representation. We estimate the autoregression on the filtered cycle, collect the fitted residuals $\hat{u}_t$, and recenter them so that the bootstrap innovations have zero mean.

For each bootstrap replication $b = 1, \ldots, B$, we draw with replacement from the recentered residuals and generate a pseudo-cycle $\{c_t^{*(b)}\}_{t=1}^T$ recursively from the estimated autoregression. We use a burn-in period so that the retained bootstrap sample has the same length as the PME window and is not sensitive to initial conditions. On each pseudo-sample we recompute the same two empirical moments,

$$m^{*(b)} = \left(\hat{\sigma}_c^{*(b)}, \hat{\rho}_c^{*(b)}(1)\right)'.$$

The aggregate activity block covariance matrix is then estimated as the empirical covariance of the bootstrap draws,

$$\widehat{\Sigma}_{agg} = \frac{1}{B-1}\sum_{b=1}^{B}\left(m^{*(b)} - \bar{m}^*\right)\left(m^{*(b)} - \bar{m}^*\right)', \tag{C3}$$

where $\bar{m}^* = B^{-1}\sum_{b=1}^{B} m^{*(b)}$.

This procedure delivers a full 2×2 covariance block for the aggregate activity moments, including the covariance between the volatility and persistence moments. It is preferable here to a purely analytical variance formula because it is aligned with the transformed cycle that actually enters the target vector and does not require reducing the empirical activity process to an AR(1) representation solely for weighting purposes.

We substitute the activity block of the original microdata-bootstrap variance-covariance matrix with the 2×2 estimates of the sieve bootstrap procedure, zeroing out the covariance terms between the aggregate activity moments and the other moments to obtain the final variance-covariance matrix $\hat{\Sigma}$.

Remark on correlations between labor and activity. The labor–activity correlations (e.g., between the unemployment rate and the cycle component of activity) depend on both cross-sectional sampling uncertainty—through the labor series—and time-series uncertainty—through the activity series. Because the survey-based labor series are substantially noisier than the published IBC-Br aggregate, the individual-level bootstrap captures the dominant source of variance for these moments. The residual underestimation from holding the activity series fixed is a second-order effect.

Final Weighting Matrix For the weighting matrix of the SMM procedure, using the full variance-covariance matrix is not desirable because the covariance terms can be poorly estimated. Our preferred alternative is to restrict $\hat{\Sigma}$ to a block-diagonal structure, where we zero out all the covariance terms but those between moments within the same block. The blocks are defined as in Table C2. Our baseline weighting matrix is then $W = (\hat{\Sigma}_{blockdiag})^{-1}$. As a robustness, we also calculate an alternative weighting matrix as the inverse of the diagonal elements of $\hat{\Sigma}$, $\tilde{W} = (\text{diag}(\hat{\Sigma}))^{-1}$.

C.3.2 Simulated Moments

For any given parameter vector P, finding the simulated moments $M_{model}(P)$ requires solving the full model and simulating it for N individuals over T periods. The simulated moments are calculated from this generated panel data using the equivalent recipes to their empirical counterparts:

- **Stock and Flow Moments**: these are the long-run averages of the simulated time series. Moments by observed skill group are recovered from the latent human-capital states using the Bayesian weights $\omega_{H,k}$ and $\omega_{L,k}$.
- **Wage Premiums**: the formal-informal wage premium is computed as the coefficient from a within-worker fixed-effects regression of log wages on a formal-sector indicator. The high-skill premium is computed as the coefficient from a pooled regression of log wages on the Bayesian high-skill weight and a formal-sector control.
- **Minimum Wage Moment**: we calculate the average wage of model-implied low-skilled, full-time, private workers from the simulation (W^L_{model}). The simulated moment is the ratio $\bar{w}_m/W^L_{model}$, where $\bar{w}_m$ is the parameter value being tested.
- **Wage Dispersion**: we calculate the standard deviation of log wages across employed workers in the simulated economy.
- **Cyclical Moments**: we apply the same HP-filter procedure to the simulated aggregate output series and labor-market aggregates after the initial burn-in period.

C.3.3 Minimization

The SMM procedure searches for the parameter vector P^* that minimizes the quadratic distance function:

$$P^* = \underset{P}{\operatorname{argmin}} \left[M_{data} - M_{model}(P)\right]' W \left[M_{data} - M_{model}(P)\right].$$

In the implemented routine, this high-dimensional optimization is performed in two stages. First, a global search routine explores the parameter space under adaptive state-space bounds. Second, a local optimizer refines the candidate solution under fixed grids. Across evaluations, the simulation uses common random numbers, and the solver can use warm starts from previous value-function iterations. The objective function also includes soft penalties in regions where the wage-premium moments become weakly identified, so that the optimizer avoids parameterizations with nearly singular identifying variation.